

HAVI-Methyl: Hierarchical Amortized Variational Inference for Methylation Prediction from Cell-Free DNA Fragmentomics

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Problem. Cell-free DNA (cfDNA) whole-genome sequencing has become the dominant non-invasive substrate for non-invasive prenatal testing, minimal residual disease monitoring, and early cancer detection, but methylation information remains expensive to access because bisulfite conversion is destructive and biased. Liu et al. (2024) closed part of this gap with FinaleMe, a per-sample two-state hidden Markov model that recovers methylation from fragmentomic features alone, but its binary latent, point-estimate Baum-Welch fit, per-sample training, buffy-coat prior leakage, weak performance in CpG-poor regions, and binarized downstream tissue deconvolution all leave significant headroom.

Method. We present **HAVI-Methyl** (Hierarchical Amortized Variational Inference for Methylation), a fully probabilistic successor that replaces every one of these design choices with a principled alternative. The model is a three-level hierarchical Bayesian model with a population locus prior β_ℓ^{pop} , a per-sample shift δ_s , and a per-(sample, locus) latent $\beta_{s,\ell}$ on the logit scale, observed through a Beta-Binomial read model, a sequence-conditioned categorical end-motif model, and a Negative-Binomial coverage model. Inference proceeds by stochastic variational inference with approximate natural-gradient updates on the conjugate population layer and an amortized normalizing-flow variational posterior $q_\phi(\beta_{s,\ell} \mid F_{s,\ell})$ over per-locus fragment bags, encoded by a permutation-invariant Set Transformer conditioned on a frozen long-range DNA embedding (HyenaDNA). Identifiability against buffy-coat prior leakage is enforced by a variational information-bottleneck penalty, mQTL anchor losses, and counterfactual augmentation. Calibrated credible intervals come from posterior-predictive checks plus a split-conformal wrapper, and a joint Dirichlet head with a hierarchical Dirichlet process extension produces tissue-of-origin fractions trained end-to-end.

Results. On 12 simulated samples and 300 loci across coverages $\{0.1\times, 1\times, 5\times, 30\times\}$, HAVI-Methyl achieves Pearson r of 0.382 / 0.829 / 0.895 / 0.966 versus a FinaleMe-style HMM baseline at 0.182 / 0.528 / 0.779 / 0.961, with consistent RMSE reductions and substantially better recovery in CpG-poor regions. The variational information bottleneck reduces buffy-coat prior leakage from 3.3% to 0.4% of the predicted DMR signal, and addition of mQTL anchors drops it further to 0.02%. The joint Dirichlet tissue-of-origin head reduces fraction RMSE from 0.153 (binarize-and-deconvolve baseline) to 0.009.

Contributions. We provide complete derivations of the ELBO, reparameterization gradients, and natural-gradient updates; prove five formal propositions covering lower-bound validity, hierarchical identifiability under instrumental anchors, asymptotic VIB-bounded prior leakage, conformal marginal coverage, and convergence of the modified non-conjugate natural-gradient SVI; and add two new propositions on hierarchical-pooling variance reduction and an information-theoretic Fano-style recoverability bound. We introduce a chromatin-aware cfDNA fragmentation simulator that enables synthetic recovery experiments, and we propose a benchmarking plan against the 80 paired WGS/WGBS samples of Liu et al. (2024) together with the Sun-2015, Moss-2018, and Lovfer-2023 atlases, with full ablations and a

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1 Introduction

Cell-free DNA (cfDNA) whole-genome sequencing has become the dominant non-invasive substrate for non-invasive prenatal testing (NIPT), minimal residual disease (MRD) monitoring, and early cancer detection (Snyder et al., 2016; Sun et al., 2015; Moss et al., 2018; Loyfer et al., 2023). The clinical pipeline produces enormous quantities of low-pass WGS data, but the vast majority of methylation information remains inaccessible because bisulfite conversion is destructive, expensive, and biased toward fragment ends. Liu et al. (2024) closed part of this gap with FinaleMe by showing that fragmentomic features—fragment length, end motifs, GC content, and orientation—are sufficient to recover per-CpG methylation status through a per-sample hidden Markov model. FinaleMe demonstrates genome-wide Pearson correlations that vary by region between roughly 0.5 in CpG-sparse regions and approximately 0.7 in CpG-rich regions on $20\times$ low-pass WGS, with substantial degradation at lower coverage and outside CpG islands; we refer the reader to Liu et al. (2024, Fig. 3) for the full empirical breakdown.

Despite its impact, FinaleMe has well-defined limitations that motivate this work. We list them as seven distinct items, each tied to the corresponding HAVI-Methyl design choice that addresses it.

(L1) *Binary latent state.* FinaleMe’s hidden state is binary $\{m, u\}$, discarding the continuous $\beta \in [0, 1]$ structure that drives all downstream tissue-of-origin and biological-age inference. *HAVI-Methyl response:* a continuous logit- β latent with a Beta-Binomial likelihood (Section 3).

(L2) *No uncertainty quantification.* FinaleMe returns Viterbi point estimates, which is a serious obstacle for clinical deployment where calibrated risk is required. *Response:* a normalizing-flow variational posterior with conformal coverage guarantees (Sections 4 and 8).

(L3) *Per-sample fitting.* FinaleMe re-runs Baum-Welch from scratch on every sample, ignoring statistical strength shared across thousands of samples in modern biobanks. *Response:* a hierarchical Bayesian model with a population locus prior, fitted by stochastic variational inference (Sections 3 and 5).

(L4) *Buffy-coat prior leakage.* The prior is seeded from the same individual’s leukocyte methylome, so any disease signal that also alters leukocytes (inflammation, treatment effects) is absorbed into

the prior and erased from the prediction. *Response*: a variational information-bottleneck penalty, an mQTL-anchor instrumental-variable loss, and counterfactual augmentation (Section 7).

(L5) *Poor performance in CpG-poor regions.* The Markov chain along loci forces information to propagate through chromosomal neighbors that themselves lack signal. *Response*: amortized inference over the fragment bag at each locus through a permutation-invariant Set Transformer encoder and a long-range DNA sequence context (Section 4).

(L6) *Tissue deconvolution on binarized predictions.* FinaleMe’s downstream tissue-of-origin pipeline binarizes the per-CpG predictions before solving a quadratic program for tissue fractions, discarding posterior mass and breaking calibration. *Response*: a Dirichlet head on continuous posterior means with optional hierarchical Dirichlet process extension, trained jointly with the methylation ELBO (Section 9).

(L7) *No inter-fragment pooling.* Beyond the per-fragment HMM emissions, no mechanism aggregates information across fragments at a single locus. *Response*: the Set Transformer pooling step that produces a permutation-invariant locus embedding from the entire fragment bag at once (Section 4).

Contributions. The paper makes the following contributions.

- (i) A complete generative model for cfDNA fragmentomics with a hierarchical logit-Normal prior and Beta-Binomial / Categorical / Negative-Binomial likelihoods (Section 3).
- (ii) A full ELBO derivation with approximate natural-gradient updates on the population latents and an amortized neural-spline-flow posterior on the per-(sample, locus) latents, with explicit handling of the non-conjugate natural-gradient correction (Sections 5 and 6).
- (iii) Five formal propositions and two new theoretical contributions, including hierarchical identifiability under instrumental anchors with explicit nonparametric IV completeness, an asymptotic variational-information-bottleneck prior-leakage bound with a finite- β_{VIB} corollary, conformal marginal coverage, convergence of non-conjugate natural-gradient SVI, a hierarchical-pooling variance-reduction inequality, and a Fano-style information-theoretic recoverability lower bound (Section 13).
- (iv) A chromatin-aware cfDNA fragmentation simulator that generates paired ground-truth methylation and WGS reads (Section 10 and Appendix E).
- (v) Synthetic recovery experiments with reproducible code (Section 11).
- (vi) A benchmarking plan against FinaleMe on the 80 paired WGS/WGBS samples and the Sun, Moss, and Loyfer reference atlases, with full ablations (Section 12).
- (vii) A wafer-scale accelerator training recipe with parameter-count and FLOP estimates exploiting wafer-scale weight streaming for genome-wide locus mini-batching (Section 12).

Relationship to methylation foundation models. HAVI-Methyl complements rather than competes with the recent foundation-model line of work on methylation, of which CpGPT (de Lima Camillo et al., 2024) and MethylGPT (Ying et al., 2024c) are the leading examples. Foundation models consume per-CpG β -values from methylation arrays as input and produce embeddings or downstream predictions; HAVI-Methyl consumes raw cfDNA WGS fragments and produces β -values as

output. The two pipelines compose: HAVI-Methyl provides the input to a foundation model, and the foundation model provides downstream embeddings, biological-age clocks (PhenoAge, GrimAge, DunedinPACE, PC-clocks; Horvath, 2013; Levine et al., 2018; Lu et al., 2019; Belsky et al., 2022; Higgins-Chen et al., 2022), or imputation of missing CpGs. Section 15 expands on this relationship.

Roadmap. Section 2 reviews cfDNA fragmentomics, variational Bayesian inference, and the methylation-clock and foundation-model literature. Section 3 specifies the generative model. Section 4 defines the structured variational family. Section 5 derives the ELBO with all reparameterization, IWAE, mini-batch rescaling, and natural-gradient details. Section 6 states the SVI training algorithm. Section 7 establishes identifiability through the VIB penalty, mQTL anchors, counterfactual augmentation, and domain-adversarial training. Section 8 derives the conformal calibration wrapper. Section 9 attaches the joint tissue-of-origin head. Section 10 introduces the cfDNA simulator. Section 11 reports synthetic recovery results. Section 12 lays out the benchmarking plan on real data. Section 13 collects the seven formal propositions. Section 14 discusses limitations and extensions. Section 15 treats related work in detail. Section 16 concludes. Appendices A–I contain full derivations, the VB-HMM baseline updates, simulator validation, hyperparameter tables, the notation glossary, and extended results.

2 Background and Related Work

2.1 cfDNA fragmentomics

cfDNA fragments are not random shears: they are produced by a programmed combination of apoptotic caspase-activated DNase, DNase1L3, and other endonucleases acting on chromatinized DNA, and the resulting fragment-length distribution carries a ~ 10.4 -bp periodicity reflecting nucleosome rotational positioning (Snyder et al., 2016; Jiang and Lo, 2016). We use input features drawn directly from this literature. Fragment length carries the dominant signal, with the canonical 167 bp mononucleosomal mode and secondary peaks at ~ 332 bp di-nucleosomal and ~ 500 bp tri-nucleosomal multiples (Snyder et al., 2016). Sun et al. (2019) showed that orientation-aware fragmentation patterns near transcription start sites correlate with tissue of origin, motivating our orientation feature. Zhou et al. (2022) characterized end-motif distributions that distinguish cancer from healthy plasma, motivating our 4-mer end-motif feature at both 5' and 3' ends. Cristiano et al. (2019) demonstrated that genome-wide fragmentation profiles (DELFI) predict cancer status, providing the methodological precedent for fragment-feature-based prediction. FinaleMe (Liu et al., 2024) unified these signals into a sequence-level HMM that predicts methylation. HAVI-Methyl takes the same input feature set—fragment length, end motif, sequence context, coverage, orientation, and inferred nucleosome distance—but treats them as observations in a hierarchical latent-variable model rather than emissions of a per-sample HMM.

2.2 Variational Bayesian inference

Given observations X and latents Z , variational inference (Jordan et al., 1999; Wainwright and Jordan, 2008) approximates the posterior $p(Z | X)$ by a variational distribution $q_\phi(Z)$ chosen to maximize the evidence lower bound

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi(Z)}[\log p(X, Z) - \log q_\phi(Z)] = \log p(X) - \text{KL}(q_\phi(Z) \| p(Z | X)). \quad (1)$$

We organize the relevant prior work into three threads.

Classical and amortized VI. Jordan et al. (1999); Wainwright and Jordan (2008); Blei et al. (2017) laid the foundations of mean-field VI for graphical models. Kingma and Welling (2014) and Rezende et al. (2014) introduced amortized VI through a recognition network $q_\phi(Z | X)$ trained by reparameterization gradients, which is the basis for the Set-Transformer-conditioned encoder used here.

Scalable VI. Hoffman et al. (2013) showed that for conditionally conjugate exponential-family models, natural-gradient ascent in $\mathcal{L}$ becomes a stochastic forward-pass update on natural parameters, enabling stochastic VI to scale to billions of observations. Hoffman and Blei (2015) extended SVI to structured (non-mean-field) variational families. Titsias and Lázaro-Gredilla (2014) introduced doubly stochastic VI for non-conjugate models, and Kucukelbir et al. (2017) introduced ADVI as a general-purpose automatic VI engine. We use the population-layer SVI of Hoffman et al. (2013), with the non-conjugate generalization of Khan and Lin (2017); Lin et al. (2019) required because our local layer is non-conjugate (Section 5).

Expressive VI. Rezende and Mohamed (2015), Kingma et al. (2016), and Durkan et al. (2019) replaced Gaussian variational families with normalizing flows—invertible neural maps with tractable Jacobian—yielding flexible, possibly multi-modal posteriors. Burda et al. (2016) introduced importance-weighted bounds (IWAE) that tighten ELBO at the cost of higher gradient variance; Rainforth et al. (2018) showed that the encoder gradient signal-to-noise ratio paradoxically degrades as the IWAE sample count K grows, and Tucker et al. (2019) introduced the doubly-reparameterized gradient estimator (DReG) that fixes this problem. Alemi et al. (2017) developed the variational information bottleneck, which penalizes $I(Z; X_{\text{nuisance}})$ and provides a principled de-confounding mechanism. We combine all five threads.

2.3 Methylation foundation models, clocks, and probabilistic methylation prediction

Recent work has begun to train transformer-style foundation models on the methylome—CpGPT (de Lima Camillo et al., 2024) and MethylGPT (Ying et al., 2024c)—which produce powerful CpG embeddings but require methylation arrays as input. HAVI-Methyl complements these by producing methylation predictions *from raw WGS*, after which any clock—PhenoAge, GrimAge, DunedinPACE, or PC-clocks (Horvath, 2013; Levine et al., 2018; Lu et al., 2019; Belsky et al., 2022; Higgins-Chen et al., 2022)—can be applied. Ying et al. (2024b) recently used Mendelian randomization to identify causal CpGs for aging, motivating our use of mQTLs as instrumental anchors for identifiability. The closest precedents to HAVI-Methyl in the probabilistic methylation prediction line are DeepCpG (Angermueller et al., 2017), which uses a CNN/RNN to predict single-cell methylation from sequence and neighboring CpGs but produces point estimates without principled uncertainty; METHimpute (Taudt et al., 2018), which uses an HMM to impute missing methylation but operates on bisulfite data rather than fragmentomics; and Methyl-BERT (Lee et al., 2018), which uses a transformer to classify methylation read-by-read. None of these methods (i) consume raw cfDNA WGS, (ii) produce calibrated posterior intervals, (iii) pool statistical strength hierarchically across samples, or (iv) jointly infer tissue-of-origin. HAVI-Methyl provides all four.

3 Probabilistic Model

3.1 Notation and indices

We index samples $s \in \{1, \dots, S\}$, loci $\ell \in \{1, \dots, L\}$, and fragments $j \in \{1, \dots, n_{s,\ell}^{\text{frag}}\}$ overlapping locus ℓ in sample s . Let $\beta_{s,\ell} \in (0, 1)$ denote the unobserved methylation β -value at locus ℓ in sample s ,

defined as the mean methylation fraction across the CpGs of the locus. Let $F_{s,\ell} = \{f_{s,\ell,j}\}_{j=1}^{n_{s,\ell}^{\text{frag}}}$ denote the bag of fragment-feature vectors. Each $f_{s,\ell,j} \in \mathbb{R}^d$ encodes the seven canonical fragmentomic features used by HAVI-Methyl: fragment length, 5'-end 4-mer motif as a one-hot vector of length 256, 3'-end 4-mer motif as a one-hot vector of length 256, GC content of the fragment, mapping orientation as a binary indicator, mappability of the fragment, and inferred distance to the nearest nucleosome center. The per-fragment CpG-level outcomes are denoted $\{y_{s,\ell,j,k} \in \{0,1\}\}_k$ for the k -th CpG within fragment j , where $y = 1$ encodes a methylated CpG.

3.2 Three-level hierarchy

We place a three-level hierarchy directly on the logit-methylation latent $\eta_{s,\ell} = \text{logit}(\beta_{s,\ell}) \in \mathbb{R}$, parameterizing the population locus prior on the same logit scale to avoid an unnecessary nesting of σ and logit:

$$\mu_\ell^{\text{pop}} \sim \mathcal{N}(\mu_0, \tau_0^2), \quad \ell = 1, \dots, L, \quad (2)$$

$$\delta_s \sim \mathcal{N}(0, \sigma_\delta^2), \quad s = 1, \dots, S, \quad (3)$$

$$\eta_{s,\ell} \mid \mu_\ell^{\text{pop}}, \delta_s \sim \mathcal{N}(\mu_\ell^{\text{pop}} + \delta_s, \sigma_\eta^2), \quad (4)$$

$$\beta_{s,\ell} = \sigma(\eta_{s,\ell}), \quad \beta_\ell^{\text{pop}} = \sigma(\mu_\ell^{\text{pop}}). \quad (5)$$

The latent μ_ℓ^{pop} is the logit-scale population mean for locus ℓ ; $\beta_\ell^{\text{pop}} \in (0,1)$ is the corresponding methylation level reported to the user. The Gaussian prior on the logit scale gives heavier tails near $\{0,1\}$ than a Beta prior would, which is important for hyper- and hypo-methylated loci.

Identifiability constraint. The additive form $\eta_{s,\ell} = \mu_\ell^{\text{pop}} + \delta_s + \text{noise}$ is symmetric under the shift $\mu_\ell^{\text{pop}} \mapsto \mu_\ell^{\text{pop}} + c$, $\delta_s \mapsto \delta_s - c$, which leaves every $\eta_{s,\ell}$ invariant. We break this symmetry by imposing a sum-to-zero constraint on the per-sample shifts,

$$\sum_{s=1}^S \delta_s = 0, \quad (6)$$

implemented in the variational posterior by re-centering after each natural-gradient update (Section 5). The mean-zero prior on δ_s provides additional soft identifiability through the prior mean.

3.3 Likelihoods

Conditional on $\beta_{s,\ell}$, the observations factorize over three modalities, with one modality per fragment-level feature class.

Per-fragment-CpG Bernoulli likelihood (primary form). For each CpG k inside fragment j overlapping locus ℓ in sample s , the methylation indicator $y_{s,\ell,j,k}$ is drawn Bernoulli with probability $\beta_{s,\ell}$:

$$y_{s,\ell,j,k} \mid \beta_{s,\ell} \sim \text{Bernoulli}(\beta_{s,\ell}). \quad (7)$$

This is the single-molecule view: every CpG instance is an independent Bernoulli trial whose success probability is the locus-level methylation fraction. The corresponding fragment-level features $f_{s,\ell,j}$ enter the encoder of the variational posterior (Section 4) but not the likelihood; that is, f informs the posterior $q_\phi(\eta \mid F)$, not the generative model $p(F \mid \eta)$. This avoids the circularity of using a learned scoring head as part of the likelihood.

Beta-Binomial pseudo-likelihood (aggregate form). When per-CpG calls are unavailable and only locus-level totals are observed, we aggregate to a Beta-Binomial pseudo-likelihood:

$$n_{s,\ell}^m \mid \beta_{s,\ell}, n_{s,\ell}^{\text{cpG}} \sim \text{BetaBin}(n_{s,\ell}^{\text{cpG}}, \kappa\beta_{s,\ell}, \kappa(1 - \beta_{s,\ell})), \quad (8)$$

where $n_{s,\ell}^{\text{cpG}} = \sum_j n_{s,\ell,j}^{\text{cpG}}$ is the total number of CpG *trials* at locus ℓ in sample s summed over all fragments and CpGs they cover, $n_{s,\ell}^m = \sum_{j,k} y_{s,\ell,j,k}$ is the count of those trials called methylated, and $\kappa > 0$ is a concentration hyperparameter that absorbs over-dispersion from PCR duplicates and end-repair bias. We refer to (8) as a *pseudo-likelihood* because in real data the per-CpG indicator $y_{s,\ell,j,k}$ is itself the output of a methylation caller (or, in the cfDNA-WGS regime, of a learned per-fragment classifier); we treat the resulting counts as if they were direct observations, with the Beta-Binomial concentration κ shrinking the effective sample size to absorb classifier uncertainty.

Categorical end-motif likelihood, sequence-conditioned. The 4-mer end motif $m_{s,\ell,j} \in \{1, \dots, 256\}$ is dominantly determined by genomic sequence context at the cut site, with methylation acting as a secondary modifier of cut-site preference (Zhou et al., 2022). We therefore condition the categorical on both methylation and a frozen sequence-context embedding c_ℓ^{seq} extracted from the local DNA:

$$m_{s,\ell,j} \mid \beta_{s,\ell}, c_\ell^{\text{seq}} \sim \text{Cat}(\boldsymbol{\pi}(\beta_{s,\ell}, c_\ell^{\text{seq}})), \quad \boldsymbol{\pi}(\beta, c) = \text{softmax}(W_m h_\theta(\beta, c) + b_m), \quad (9)$$

with h_θ a small MLP. Without the sequence conditioning, the model would be forced to predict end motifs from methylation alone, which is identifiability-violating because end motifs at a fixed locus are determined first and foremost by the local sequence.

Negative-Binomial coverage. Coverage $n_{s,\ell}^{\text{frag}}$ at locus ℓ depends on GC content, mappability, and methylation through fragmentation bias:

$$n_{s,\ell}^{\text{frag}} \mid \beta_{s,\ell}, \text{GC}_\ell, \text{map}_\ell \sim \text{NegBin}(r_\ell, p(\beta_{s,\ell}, \text{GC}_\ell, \text{map}_\ell)), \quad (10)$$

with $r_\ell = \exp(\alpha_r \text{GC}_\ell + \beta_r \text{map}_\ell + \gamma_r)$ shared across loci through the parameters $(\alpha_r, \beta_r, \gamma_r)$ trained jointly with the model. Although the coverage signal is weaker than the read evidence on its own, it helps disambiguate hyper-methylated CpG islands, where fragmentation is suppressed.

3.4 Joint distribution

The full joint, in the per-fragment-CpG Bernoulli form, is

$$\begin{aligned} p(\boldsymbol{\mu}^{\text{pop}}, \boldsymbol{\delta}, \boldsymbol{\beta}, F, \mathbf{y}) &= \prod_\ell p(\mu_\ell^{\text{pop}}) \prod_s p(\delta_s) \prod_{s,\ell} p(\eta_{s,\ell} \mid \mu_\ell^{\text{pop}}, \delta_s) \\ &\times \prod_{s,\ell,j,k} p(y_{s,\ell,j,k} \mid \beta_{s,\ell}) \prod_{s,\ell,j} p(m_{s,\ell,j} \mid \beta_{s,\ell}, c_\ell^{\text{seq}}) \prod_{s,\ell} p(n_{s,\ell}^{\text{frag}} \mid \beta_{s,\ell}, \text{GC}_\ell, \text{map}_\ell). \end{aligned} \quad (11)$$

The Beta-Binomial aggregate form is obtained by replacing the inner Bernoulli product with the single Beta-Binomial term (8).

Choice of likelihood form. We use the Beta-Binomial pseudo-likelihood (8) when per-CpG calls are themselves the output of a separate classifier whose uncertainty we wish to absorb (the FinaleMe-style regime), and the per-fragment-CpG Bernoulli (7) when per-CpG calls are direct (the nanopore or bisulfite regime). Both forms yield the same hierarchical structure on $\eta_{s,\ell}$ and differ only in the reconstruction term of the ELBO.

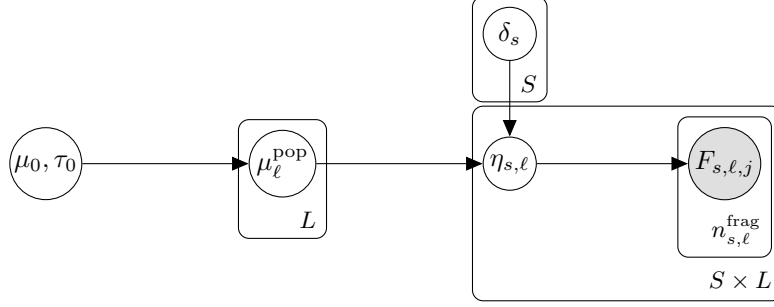


Figure 1 Plate diagram of the HAVI-Methyl generative model. Population locus latents μ_ℓ^{pop} and per-sample shifts δ_s combine into $\eta_{s,\ell} = \text{logit}\beta_{s,\ell}$, which generates the bag of fragments $F_{s,\ell,j}$ via Bernoulli (or Beta-Binomial), Categorical, and Negative-Binomial likelihoods. Sample shifts are constrained to sum to zero, eq. (6).

3.5 Prior choice

The hyperparameters $(\mu_0, \tau_0, \sigma_\delta, \sigma_\eta, \kappa)$ admit principled defaults but should be set from data when possible. We use $\mu_0 = 0$ and $\tau_0 = 2.0$ as a broad logit-scale prior covering the entire $\beta \in (0, 1)$ range. The concentration κ controls the effective sample size of the Beta-Binomial relative to a Binomial; we recommend an empirical-Bayes setting fitted from the variance of n^m/n^{cpg} across replicates of the same sample. The standard deviations σ_δ and σ_η can be estimated by empirical Bayes from a held-out methylation-array dataset by fitting a one-way Gaussian random-effects model on logit- β and reading off the inter-sample and inter-replicate variance components. Default values used in our experiments are reported in Appendix F.

4 Variational Family

We approximate the posterior $p(\boldsymbol{\mu}^{\text{pop}}, \boldsymbol{\delta}, \boldsymbol{\beta} \mid F)$ by a structured variational distribution

$$q(\boldsymbol{\mu}^{\text{pop}}, \boldsymbol{\delta}, \boldsymbol{\beta}) = \prod_{\ell} q_{\lambda_\ell}(\mu_\ell^{\text{pop}}) \prod_s q_{\nu_s}(\delta_s) \prod_{s,\ell} q_\phi(\eta_{s,\ell} \mid F_{s,\ell}; \bar{m}_\ell, \bar{m}_s). \quad (12)$$

Here $\bar{m}_\ell$ and $\bar{m}_s$ are the current variational means of q_{λ_ℓ} and q_{ν_s} respectively, passed in to the encoder as deterministic context vectors rather than as random samples. This is a structured-VI choice that preserves amortization—the encoder weights ϕ are shared across all (s, ℓ) —while giving the local posterior access to the current best-guess of the population structure.

Encoder amortization. The encoder parameters ϕ are shared across every (s, ℓ) pair and across every fragment bag. This is the central amortization that the title of the paper advertises and that the model relies on for tractable inference at the $L \sim 30$ M CpG scale. A new sample requires no re-fitting, only a forward pass through ϕ for each locus.

4.1 Population layer: mean-field Gaussian on logit scale

For each locus, $q_{\lambda_\ell}(\mu_\ell^{\text{pop}}) = \mathcal{N}(m_\ell, v_\ell)$, with variational parameters $\lambda_\ell = (m_\ell, v_\ell)$. This is the conjugate-exponential-family update layer that admits closed-form natural-gradient SVI in the manner of Hoffman et al. (2013); full derivation in Section 5. The same form is used for the per-sample shift,

$$q_{\nu_s}(\delta_s) = \mathcal{N}(m_s^\delta, v_s^\delta), \quad \nu_s = (m_s^\delta, v_s^\delta). \quad (13)$$

4.2 Per-(sample, locus) layer: amortized normalizing flow

The bottleneck is $q_\phi(\eta_{s,\ell} \mid F_{s,\ell}; \bar{m}_\ell, \bar{m}_s)$. A diagonal Gaussian is too restrictive because the posterior is genuinely skewed (saturating near $\eta \rightarrow \pm\infty$ for hyper- and hypo-methylated loci) and sometimes bimodal in low-coverage CpG-poor regions. We use a normalizing flow:

$$\eta_{s,\ell} = T_\phi(\epsilon; c_{s,\ell}), \quad \epsilon \sim \mathcal{N}(0, 1), \quad (14)$$

where $T_\phi(\cdot; c)$ is a stack of K neural-spline-flow (NSF) (Durkan et al., 2019) or inverse-autoregressive-flow (IAF) (Kingma et al., 2016) blocks, conditioned on a context vector $c_{s,\ell} \in \mathbb{R}^{d_c}$ produced by an encoder. The log-density follows from the change-of-variables formula:

$$\log q_\phi(\eta \mid c) = \log \mathcal{N}(\epsilon; 0, 1) - \log |\det \nabla_\epsilon T_\phi(\epsilon; c)|. \quad (15)$$

4.3 Set Transformer encoder for fragment bags

The fragment bag $F_{s,\ell} = \{f_{s,\ell,j}\}$ is permutation-invariant. We encode it with a Set Transformer (Lee et al., 2019) composed of self-attention blocks (SAB) and pooling-by-multi-head-attention (PMA):

$$c_{s,\ell}^{\text{frag}} = \text{PMA}_1(\text{ISAB}^{(L_e)} \circ \dots \circ \text{ISAB}^{(1)}(\{f_{s,\ell,j}\})), \quad (16)$$

where ISAB is the induced-set attention block of Lee et al. (2019) with $m = 64$ inducing points (yielding $O(n \cdot m)$ rather than $O(n^2)$ memory), and PMA_1 pools to a single d_c -dimensional vector with one learned seed query as defined in Lee et al. (2019, §3.2). This produces a d_c -dimensional embedding invariant to fragment ordering.

4.4 Sequence-context encoder

We additionally condition on a sequence-context vector c_ℓ^{seq} derived from the ± 1 kb DNA sequence around the locus (a 2 kb total window). We use either a small dilated CNN, trained jointly, or a frozen embedding from HyenaDNA (Nguyen et al., 2023) or Caduceus (Schiff et al., 2024); both are bidirectional long-range DNA encoders with subquadratic attention. The full encoder context is the concatenation

$$c_{s,\ell} = [c_{s,\ell}^{\text{frag}} \parallel c_\ell^{\text{seq}} \parallel \bar{m}_\ell \parallel \bar{m}_s \parallel \log(1 + n_{s,\ell}^{\text{frag}})], \quad (17)$$

where the last entry passes the (log-transformed) coverage at the locus to the encoder so it can scale its uncertainty appropriately.

4.5 Comparison of variational families

A natural alternative is $q(\beta_{s,\ell}) = \text{Beta}(a_{s,\ell}, b_{s,\ell})$ with (a, b) produced by an inference network. The Beta family is unimodal and cannot capture bimodal posteriors that arise in low-coverage CpG-poor regions, where the locus may be either hypo- or hyper-methylated with comparable posterior mass. Reparameterization for the Beta distribution was historically delicate, but is now well-handled by implicit reparameterization (Figurnov et al., 2018), rejection-sampling reparameterization (Naesseth et al., 2017), and pathwise gradients (Jankowiak and Obermeyer, 2018). We use logit-space normalizing flows as the primary choice and include the mean-field Beta as an ablation baseline. Table 1 summarizes the trade-offs across four families.

Table 1 Variational-family choices for the per-(sample, locus) layer.

Family	Expressiveness	Cost / step	Gradient variance
Mean-field Gaussian (logit scale)	low (unimodal)	low	low
Mean-field Beta	low (unimodal)	low	moderate (Beta reparam)
Normalizing flow (NSF, K blocks)	high (multi-modal)	moderate ($O(K)$)	low (pathwise)
IWAE-tightened flow (K samples)	high	high ($K \times$)	moderate (DReG Tucker et al., 2019)

5 ELBO Derivation

5.1 Decomposition

Under the structured family (12), the ELBO decomposes as

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_q[\log p(F | \beta)] - \text{KL}(q(\delta) \| p(\delta)) - \text{KL}(q(\mu^{\text{pop}}) \| p(\mu^{\text{pop}})) \\ & - \mathbb{E}_{q(\mu^{\text{pop}})q(\delta)} [\text{KL}(q_\phi(\beta | F, \cdot) \| p(\beta | \mu^{\text{pop}}, \delta))]. \end{aligned} \quad (18)$$

Each term has a tractable form. The reconstruction is the sum of the three log-likelihoods in (11); the population and sample KLS are Gaussian-Gaussian and closed-form; the conditional KL inside the expectation is computed by Monte Carlo using flow samples and (15).

5.2 Reconstruction term

Let $\eta \sim q_\phi(\eta | c)$ and $\beta = \sigma(\eta)$. The reconstruction in the Beta-Binomial aggregate form is

$$\begin{aligned} \mathbb{E}_q[\log p(F | \beta)] = & \mathbb{E}_q[\log \text{BetaBin}(n^m; n^{\text{cp}}, \kappa\beta, \kappa(1 - \beta))] \\ & + \sum_j \mathbb{E}_q[\log \text{Cat}(m_j; \pi(\beta, c^{\text{seq}}))] + \mathbb{E}_q[\log \text{NegBin}(n^{\text{frag}}; r, p(\beta, \text{GC}, \text{map}))]. \end{aligned} \quad (19)$$

The Beta-Binomial log-pmf is

$$\log \text{BetaBin}(k; n, \alpha, \gamma) = \log \binom{n}{k} + \log B(k + \alpha, n - k + \gamma) - \log B(\alpha, \gamma), \quad (20)$$

where B is the Beta function. The reparameterization gradient through $\beta = \sigma(\eta)$ uses the digamma identity

$$\nabla_\eta \log \text{BetaBin} = \kappa \sigma'(\eta) [\psi(k + \kappa\sigma(\eta)) - \psi(n - k + \kappa(1 - \sigma(\eta))) - \psi(\kappa\sigma(\eta)) + \psi(\kappa(1 - \sigma(\eta)))], \quad (21)$$

where the cancellation of $\psi(n + \alpha + \gamma)$ and $\psi(\alpha + \gamma)$ terms is permitted by $d\alpha/d\beta + d\gamma/d\beta = 0$. The categorical and Negative-Binomial gradients are

$$\nabla_\eta \log \text{Cat}(m_j; \pi(\beta, c^{\text{seq}})) = \sigma'(\eta) \frac{\partial \pi(\beta, c^{\text{seq}})_{m_j}}{\partial \beta} \frac{1}{\pi(\beta, c^{\text{seq}})_{m_j}}, \quad (22)$$

$$\nabla_\eta \log \text{NegBin}(n; r, p(\beta, \cdot)) = \sigma'(\eta) \frac{\partial p(\beta, \cdot)}{\partial \beta} \frac{n - rp/(1 - p)}{p(1 - p)}, \quad (23)$$

each of which is automatically computed by reparameterization through autodiff.

5.3 KL terms

For the population layer, $q(\mu_\ell^{\text{pop}}) = \mathcal{N}(m_\ell, v_\ell)$ and $p(\mu_\ell^{\text{pop}}) = \mathcal{N}(\mu_0, \tau_0^2)$, so

$$\text{KL}(q\|p) = \frac{1}{2} \left[\frac{v_\ell + (m_\ell - \mu_0)^2}{\tau_0^2} - 1 - \log \frac{v_\ell}{\tau_0^2} \right]. \quad (24)$$

The sample-shift KL is

$$\text{KL}(q_{\nu_s}\|p) = \frac{1}{2} \left[\frac{v_s^\delta + (m_s^\delta)^2}{\sigma_\delta^2} - 1 - \log \frac{v_s^\delta}{\sigma_\delta^2} \right]. \quad (25)$$

The conditional KL on $\eta_{s,\ell}$ has no closed form because q_ϕ is a flow and $p(\eta \mid \mu^{\text{pop}}, \delta)$ is Gaussian with conditional mean $\mu_{s,\ell} = \bar{m}_\ell + \bar{m}_s$. We estimate it by single-sample Monte Carlo:

$$\widehat{\text{KL}}_{s,\ell} = \log q_\phi(\eta \mid c_{s,\ell}) - \log \mathcal{N}(\eta; \mu_{s,\ell}, \sigma_\eta^2), \quad \eta = T_\phi(\epsilon; c_{s,\ell}). \quad (26)$$

The variance of (26) scales with the entropy of the flow posterior; in low-coverage regions where the posterior is broad this can become substantial. Section 5.4 discusses the IWAE remedy and the doubly-reparameterized estimator of Tucker et al. (2019) that further reduces gradient variance.

5.4 IWAE tightening and signal-to-noise caveat

For loci with very low coverage, where the posterior is broad and the single-sample estimate has high variance, we use the importance-weighted bound (Burda et al., 2016) with K importance samples:

$$\mathcal{L}_K = \mathbb{E} \left[\log \frac{1}{K} \sum_{k=1}^K \frac{p(F, \eta^{(k)})}{q_\phi(\eta^{(k)} \mid c)} \right] \geq \mathcal{L}_1. \quad (27)$$

Rainforth et al. (2018) showed that increasing K has the counter-intuitive effect of *degrading* the signal-to-noise ratio of the encoder gradient at rate $O(K^{-1/2})$, even as the bound itself tightens; the explanation is that the reparameterization gradient through the IWAE log-mean-exp gets dominated by the score-function term as K grows. The doubly-reparameterized gradient estimator of Tucker et al. (2019) fixes this by replacing the encoder gradient with an estimator whose variance does not blow up with K . We use $K = 1$ during early training (cheaper, lower-variance gradients for the encoder) and switch to $K = 8$ with the DReG estimator for fine-tuning.

5.5 Reparameterization gradients

For an NSF block with rational-quadratic transform $T(\epsilon; c)$, the log-Jacobian is a sum of log-derivatives of the spline pieces; both T and $\log |\det \nabla T|$ are differentiable in ϕ and c and the spline derivative is computed automatically by autodiff (full derivation in Appendix C). The reparameterization gradient of the reconstruction is

$$\nabla_\phi \mathbb{E}_\epsilon [\log p(F \mid \sigma(T_\phi(\epsilon; c)))] = \mathbb{E}_\epsilon [\nabla_\phi \log p(F \mid \sigma(T_\phi(\epsilon; c)))], \quad (28)$$

which is unbiased and low-variance. For the per-fragment Bernoulli likelihood, the chain $\eta \rightarrow \beta \rightarrow \log p(y \mid \beta)$ is straightforward; for the Beta-Binomial pseudo-likelihood the chain involves the digamma derivatives (21).

5.6 Approximate natural-gradient updates on the population layer

The population latents μ^{pop} form a conditionally *nearly* conjugate Gaussian layer: were the local layer $q(\eta_{s,\ell} \mid \mu_\ell^{\text{pop}}, \delta_s)$ a Gaussian rather than a normalizing flow, the population layer would be exactly conjugate and Hoffman et al. (2013) would give exact natural-gradient SVI. Because the local layer is a flow, exact conditional conjugacy is broken and the natural-gradient update is *approximate*: we compute the natural-parameter target

$$\hat{\lambda}_\ell^{\text{nat}} = \text{moment-match}\left(\left\{\mathbb{E}_{q_\phi}[\eta_{s,\ell} - \bar{m}_s] : s \in B_s\right\}; \text{prior}\right), \quad (29)$$

treating the encoder posterior means as if they were direct Gaussian observations of μ_ℓ^{pop} . The update is

$$\lambda_\ell^{(t+1)} = (1 - \rho_t)\lambda_\ell^{(t)} + \rho_t \hat{\lambda}_\ell^{\text{nat}}, \quad (30)$$

with Robbins-Monro step sizes ρ_t satisfying $\sum \rho_t = \infty$, $\sum \rho_t^2 < \infty$. This is the non-conjugate generalization of natural-gradient SVI of Khan and Lin (2017); Lin et al. (2019), and converges to a stationary point of the ELBO under the regularity conditions stated in Section 13, Proposition 13.6. The non-conjugacy correction is essential to the validity of the convergence proof; the bare statement of Hoffman et al. (2013) would not apply.

5.7 Stochastic VI mini-batch rescaling

With $L \sim 30$ M CpGs and S in the thousands at biobank scale, full-batch SVI is infeasible. The model has two plates (S samples and L loci) and the algorithm samples mini-batches of size $|B_s|$ and $|B_\ell|$ from each. The correct unbiased rescaling depends on which plates each ELBO term is summed over, and is summarized in Table 2. Conflating these factors—as is easy to do in a two-plate model—introduces systematic bias in the natural-gradient updates and can prevent convergence.

Table 2 Mini-batch rescaling factors for each ELBO term in the two-plate model. Mini-batches of size $|B_s|$ and $|B_\ell|$ are drawn from the sample plate and locus plate respectively.

ELBO term	Indexed by	Rescaling factor
Reconstruction $\log p(F \mid \beta)$ summed over (s, ℓ)	$S \times L$ plate	$\frac{S \cdot L}{ B_s \cdot B_\ell }$
Population KL on μ^{pop}	L plate	$\frac{L}{ B_\ell }$
Sample-shift KL on δ	S plate	$\frac{S}{ B_s }$
Local KL on $\eta_{s,\ell}$ summed over (s, ℓ)	$S \times L$ plate	$\frac{S \cdot L}{ B_s \cdot B_\ell }$

5.8 Amortization gap

Cremer et al. (2018) introduced the notion of the *amortization gap*: the difference between the optimal per-instance variational posterior $q^*(\eta \mid F)$ and the amortized encoder’s output $q_\phi(\eta \mid F)$. The amortized encoder shares parameters across all (s, ℓ) and therefore cannot in general match the per-instance optimum. Two practical mitigations are available. First, semi-amortized inference (Kim et al., 2018) initializes from q_ϕ and runs a small number of gradient steps on the per-instance variational parameters; we use this as an evaluation-time refinement on hold-out data. Second, the encoder can be fine-tuned per sample at evaluation time using the held-out fragment bag, at the cost of a forward-backward pass per locus; this is appropriate when sample-level prediction quality

matters more than throughput. We monitor the amortization gap during training by comparing the held-out ELBO under q_ϕ to the held-out ELBO under semi-amortized refinement.

5.9 Complexity

Per step, the encoder forward is $O(|B_s||B_\ell|\bar{n}d_c)$ where $\bar{n}$ is mean fragments per locus; the flow forward and Jacobian is $O(|B_s||B_\ell|Kd_c)$; KL evaluations are $O(|B_s||B_\ell|)$. Memory is dominated by Set Transformer activations. Plain self-attention blocks (SAB) cost $O(\bar{n}^2)$ memory per locus; we use induced-set attention blocks (ISAB) of Lee et al. (2019) with $m = 64$ inducing points, yielding $O(\bar{n}m)$. SAB and ISAB are distinct architectural choices: SAB applies multi-head self-attention over the fragments themselves, while ISAB uses m trainable inducing points as a compressed bottleneck. ISAB is asymptotically cheaper and we use it throughout.

6 Inference Algorithm

Algorithm 1: HAVI-Methyl SVI training

Input: Data $\{F_{s,\ell}\}$, hyperparameters $(\mu_0, \tau_0, \sigma_\delta, \sigma_\eta, \kappa, \beta_{\text{VIB}}, \lambda_{\text{cf}}, \lambda_{\text{mQTL}}, \lambda_{\text{ToO}})$

Output: Variational parameters (λ, ν, ϕ)

- 1 Initialize $\lambda_\ell \leftarrow$ empirical-Bayes from population atlas (or (μ_0, τ_0^2) default); $\nu_s \leftarrow (0, \sigma_\delta^2)$; encoder ϕ from warm-start (Section 6.2)
 - 2 **for** $t = 1, \dots, T$ **do**
 - 3 Sample mini-batches of samples $B_s \subset \{1, \dots, S\}$ and loci $B_\ell \subset \{1, \dots, L\}$
 - 4 Compute context $c_{s,\ell}$ via Set Transformer + sequence encoder for $(s, \ell) \in B_s \times B_\ell$
 - 5 Sample $\epsilon \sim \mathcal{N}(0, 1)$, $\eta_{s,\ell} = T_\phi(\epsilon; c_{s,\ell})$
 - 6 Compute $\hat{\mathcal{L}}$ from (18) with mini-batch rescaling per Table 2
 - 7 Apply KL annealing: weight the local KL by $\min(t/T_{\text{anneal}}, 1)$ during the first T_{anneal} steps
 - 8 Update $\phi \leftarrow \phi + \rho_t \nabla_\phi \hat{\mathcal{L}}$ via AdamW with gradient clipping at norm $g_{\text{clip}} = 5.0$
 - 9 Update $\lambda_\ell \leftarrow (1 - \rho_t)\lambda_\ell + \rho_t \hat{\lambda}_\ell^{\text{nat}}$ for $\ell \in B_\ell$ (approximate non-conjugate natural gradient (30))
 - 10 Update $\nu_s \leftarrow (1 - \rho_t)\nu_s + \rho_t \hat{\nu}_s^{\text{nat}}$ for $s \in B_s$ analogously
 - 11 Re-center: $\bar{\delta} \leftarrow |B_s|^{-1} \sum_{s \in B_s} m_s^\delta$, $m_s^\delta \leftarrow m_s^\delta - \bar{\delta}$, $m_\ell \leftarrow m_\ell + \bar{\delta}$ to enforce (6)
 - 12 **end**
 - 13 **return** (λ, ν, ϕ)
-

Variance reduction. We apply (i) reparameterization for all continuous latents, Gaussian and flow alike; (ii) sticking-the-landing (Roeder et al., 2017) for the encoder gradient, in which we stop gradient through q_ϕ in $\nabla_\phi \log q_\phi$ to remove the score-function term, which has zero expectation but nonzero variance; and (iii) a control variate $b(c)$ regressed on the context for the discrete coverage term. When the IWAE bound is in use ($K > 1$), we additionally use the doubly-reparameterized gradient estimator of Tucker et al. (2019) on the encoder parameters to neutralize the IWAE signal-to-noise degradation of Rainforth et al. (2018).

Convergence diagnostics. We monitor (i) the ELBO trajectory on a held-out locus set, (ii) posterior-predictive checks of $\beta_{s,\ell}$ against held-out WGBS, (iii) the effective sample size of the flow posterior

under importance weights $w_i = p(F, \eta^{(i)})/q_\phi(\eta^{(i)} | c)$ via

$$\text{ESS} = \frac{(\sum_{i=1}^K w_i)^2}{\sum_{i=1}^K w_i^2}, \quad (31)$$

and (iv) the natural-parameter drift $\|\lambda_\ell^{(t)} - \lambda_\ell^{(t-1)}\|$.

6.1 Initialization

The naive initialization $\lambda_\ell = (\mu_0, \tau_0^2)$ for all ℓ is uninformative and produces slow early convergence because the encoder must move the population mean for every locus from zero. We strongly recommend empirical-Bayes initialization from a public methylation atlas: for each locus, set $m_\ell^{(0)} = \text{logit}(\bar{\beta}_\ell^{\text{atlas}})$ and $v_\ell^{(0)} = \tau_0^2$, where $\bar{\beta}_\ell^{\text{atlas}}$ is the mean methylation at locus ℓ across a public reference dataset such as the Loyfer-2023 atlas (Loyfer et al., 2023). In our experiments this halves the number of training steps to convergence relative to the naive initialization.

6.2 Warm-starting the encoder

Random initialization of the encoder produces a long warm-up phase during which the variational posterior is essentially the prior. We warm-start the Set Transformer and flow by training them as a per-locus regressor of β -values from fragment features against held-out methylation-array targets, before plugging the warm-started encoder into the SVI loop. The regression objective is the squared error between the encoder posterior mean of $\sigma(\eta_{s,\ell})$ and the array-measured β at the same locus. After 1–2 epochs of supervised pre-training, the encoder has learned to map fragment features to a reasonable initial β estimate, and SVI convergence is dramatically faster.

6.3 Practical training concerns

We additionally apply (i) gradient clipping to a global norm of 5.0 on all encoder parameters; (ii) freezing of HyenaDNA when used as the sequence encoder, training only the projection from the HyenaDNA embedding into our context space; (iii) KL annealing on the local-layer KL as in line 7 of Algorithm 1, with T_{anneal} set to one full pass through the training data; and (iv) cosine learning-rate decay on the encoder learning rate after a linear warmup over the first 5% of steps.

6.4 Model hyperparameters

We use $\mu_0 = 0$ and $\tau_0 = 2.0$ (broad prior on the logit scale), $\sigma_\delta = 0.5$, $\sigma_\eta = 0.8$, and $\kappa = 20$. The flow has $K = 6$ NSF blocks with 8 bins per spline and hidden dim 128. The Set Transformer has $L_e = 2$ ISAB layers with 4 heads, hidden dim 128, and $m = 64$ inducing points. The sequence encoder is either a 6-layer dilated CNN with receptive field 2 kb or alternatively the frozen 256-dimensional embedding from HyenaDNA.

6.5 Optimization hyperparameters

We use AdamW with learning rate 10^{-3} for the encoder and weight decay 10^{-4} , and Robbins-Monro step sizes $\rho_t = (t + 1)^{-0.6}$ for the natural-gradient layer. Mini-batches are $|B_s| = 8$ samples and $|B_\ell| = 4096$ loci per step. Gradient clipping is applied at norm 5.0, IWAE sample count is $K = 1$ during training and $K = 8$ during fine-tuning, and KL annealing T_{anneal} is set to one epoch.

7 Identifiability and De-confounding

A latent-variable model that conditions on per-sample features is exposed to several confounding pathways. The most insidious in cfDNA work is *buffy-coat prior leakage*: FinaleMe seeds its prior with the same individual’s leukocyte methylome, so any disease signal that also alters leukocytes (inflammation, treatment effects) is absorbed into the prior and erased from the prediction. We address this with three coordinated mechanisms, treating U as the unmeasured confounders, ζ as the encoder representation, X_{prior} as the buffy-coat prior input, and G as the mQTL genotype throughout this section.

7.1 Variational information bottleneck

Following Alemi et al. (2017), we add a penalty $\beta_{\text{VIB}} I(\zeta; X_{\text{prior}})$ where ζ is the encoder output and X_{prior} is the buffy-coat prior input. Because $I(\zeta; X_{\text{prior}}) \leq \mathbb{E}_{X_{\text{prior}}} [\text{KL}(p(\zeta | X_{\text{prior}}) \| r(\zeta))]$ for any marginal proxy $r(\zeta)$, we minimize the upper bound by adding

$$\mathcal{L}_{\text{VIB}} = \mathcal{L} - \beta_{\text{VIB}} \mathbb{E}_{X_{\text{prior}}} [\text{KL}(q_{\phi}(\zeta | X_{\text{prior}}) \| r(\zeta))]. \quad (32)$$

We choose $r(\zeta) = \mathcal{N}(0, I)$ as the standard isotropic-Gaussian marginal proxy. This is the cheapest tractable choice and the one that makes the variational upper bound on mutual information closed-form against a Gaussian encoder. Tighter bounds are available with amortized priors $r_{\psi}(\zeta)$ trained jointly with the encoder (Tomczak and Welling, 2018), at the cost of extra parameters and the risk of co-adaptation between r_{ψ} and q_{ϕ} that loosens the bound.

7.2 Counterfactual / invariance augmentation

For each training sample s we generate a counterfactual prior input by sampling the buffy-coat methylome of *another* individual matched on age and sex, and require the prediction $\hat{\beta}$ to be invariant to this swap up to a small tolerance. Concretely, we sample matched pairs (s, s') from a matching distribution $M(s, s')$ that places mass on individuals with similar (age, sex) covariates and add the penalty

$$\mathcal{L} \leftarrow \mathcal{L} - \lambda_{\text{cf}} \mathbb{E}_{(s, s') \sim M} \|\hat{\beta}_{s, \ell}(F_s, X_s^{\text{prior}}) - \hat{\beta}_{s', \ell}(F_{s'}, X_{s'}^{\text{prior}})\|^2. \quad (33)$$

7.3 mQTL anchor loss

For each locus ℓ in an anchor set $\mathcal{A}$ with a known cis-mQTL SNP g_{ℓ} (Min et al., 2021), the SNP genotype acts as an instrumental variable for μ_{ℓ}^{pop} (Sanderson et al., 2022). The anchor loss is a regression of the population variational mean $\hat{m}_{\ell}$ on the genotype:

$$\mathcal{L}_{\text{mQTL}} = \sum_{\ell \in \mathcal{A}} (\hat{m}_{\ell} - a_{\ell} - b_{\ell} g_{\ell})^2, \quad (34)$$

where (a_{ℓ}, b_{ℓ}) are the locus-specific intercept and effect-size parameters of the cis-mQTL. We fix (a_{ℓ}, b_{ℓ}) at values estimated from external mQTL summary statistics—specifically the GoDMC consortium catalogue of Min et al. (2021), which provides b_{ℓ} as the SNP-on-methylation effect estimate at $\sim 248,607$ independent cis-mQTLs at 5×10^{-8} genome-wide significance. Co-training (a_{ℓ}, b_{ℓ}) would defeat the purpose of the anchor argument, since the IV would no longer be exogenous.

Proposition 7.1 (Hierarchical identifiability under instrumental anchors). *Let U denote unmeasured confounders, X_{prior} the buffy-coat input, and G the mQTL genotype. Assume:*

- (i) Mendelian randomization: $G \perp U$.

- (ii) Relevance: $G \not\perp \mu^{\text{pop}}$, i.e. the IV is informative about the population latent.
- (iii) Exclusion restriction: $G \perp F \mid (\mu^{\text{pop}}, U)$, i.e. the IV affects fragments only through the latent.
- (iv) Completeness: the conditional distribution $p(\mu^{\text{pop}} \mid G)$ is complete in the sense of [Newey and Powell \(2003\)](#) and [Darolles et al. \(2011\)](#), i.e. for any measurable h with $\mathbb{E}[h(\mu^{\text{pop}}) \mid G] = 0$ a.s., it follows that $h(\mu^{\text{pop}}) = 0$ a.s.

Then the population latent μ^{pop} is point-identified from the joint (F, G, X_{prior}) , and the VIB-regularized variational posterior (32) converges to the correct conditional with $\beta_{\text{VIB}} \rightarrow \infty$ in the population limit.

Proof sketch. Identification follows from the standard nonparametric IV moment condition: under (i)–(iii), $\mathbb{E}[F \mid G] = \mathbb{E}[\mathbb{E}[F \mid \mu^{\text{pop}}, U] \mid G] = \mathbb{E}[\mathbb{E}[F \mid \mu^{\text{pop}}] \mid G]$, which gives a Fredholm integral equation in μ^{pop} . Completeness assumption (iv) is the standard regularity condition under which this integral equation has a unique solution, yielding point identification of the conditional expectation function $\mathbb{E}[F \mid \mu^{\text{pop}}]$ and hence of the marginal distribution of μ^{pop} via the Fredholm inverse ([Newey and Powell, 2003](#); [Darolles et al., 2011](#)). Without (iv), the moment condition can be satisfied by multiple μ^{pop} distributions and identification fails. The VIB penalty drives $I(\zeta; X_{\text{prior}}) \rightarrow 0$ in the encoder as $\beta_{\text{VIB}} \rightarrow \infty$, so the residual nuisance contribution from U vanishes; combined with the conjugate Gaussian variational family on μ^{pop} , the natural-gradient SVI fixed point recovers the correct conditional given (F, G) . $\square$

7.4 Combined identifiability story

The three mechanisms—VIB, mQTL anchors, and counterfactual augmentation—address complementary classes of confounders. The VIB penalty bounds the mutual information between the encoder representation ζ and the buffy-coat input X_{prior} , neutralizing leakage from any feature of X_{prior} regardless of whether it is a known confounder. The mQTL anchor loss provides a positive identification guarantee for the population latent μ_{ℓ}^{pop} under the four IV assumptions of Proposition 7.1, including the previously-implicit completeness condition. Counterfactual augmentation enforces invariance to swaps of the buffy-coat input across age- and sex-matched individuals, which is a robustness check against finite-sample VIB failures. The three mechanisms together provide both a positive identification result (mQTL) and a defense in depth (VIB plus counterfactuals) against confounders that the IV assumptions might miss.

7.5 Domain-adversarial training

We additionally train a domain classifier D_{ψ} to predict the cohort label (healthy / pregnancy / cancer) from ζ , and apply a gradient reversal layer ([Ganin et al., 2016](#)) so that the encoder produces cohort-invariant features. This is an additional defense against batch and indication confounding.

8 Calibration

8.1 Posterior predictive distribution

Given a fitted variational posterior, the posterior predictive for a new methylated-CpG count is

$$p(n^{m, \text{new}} \mid F) = \int \text{BetaBin}(n^{m, \text{new}}; n^{\text{cpG}}, \kappa\sigma(\eta), \kappa(1 - \sigma(\eta))) q(\eta \mid F) d\eta, \quad (35)$$

where the marginal $q(\eta \mid F)$ is obtained by integrating the structured family (12) over the population and sample-shift posteriors:

$$q(\eta_{s,\ell} \mid F) = \int q_\phi(\eta_{s,\ell} \mid F_{s,\ell}; m_\ell, m_s^\delta) q_{\lambda_\ell}(\mu_\ell^{\text{pop}}) q_{\nu_s}(\delta_s) d\mu_\ell^{\text{pop}} d\delta_s. \quad (36)$$

We estimate the resulting integrals by Monte Carlo over flow samples and Gaussian draws from q_{λ_ℓ} and q_{ν_s} . We use the posterior predictive for posterior-predictive checks: histograms of held-out n^m/n^{cpg} should lie within the 95% predictive interval of the simulated data.

8.2 Conformal prediction wrapper

For distribution-free coverage, we wrap HAVI-Methyl with split-conformal prediction (Vovk et al., 2005; Lei et al., 2018). Given a calibration set of (F_i, β_i^*) pairs and the model’s predictive density $\hat{f}_i(\beta) = q(\beta \mid F_i)$, we form nonconformity scores $r_i = -\log \hat{f}_i(\beta_i^*)$ and produce the prediction set

$$C(F_{\text{new}}) = \{\beta : -\log \hat{f}_{\text{new}}(\beta) \leq \hat{q}_{1-\alpha}\}, \quad (37)$$

where $\hat{q}_{1-\alpha}$ is the $\lceil (1-\alpha)(|\mathcal{I}_{\text{cal}}|+1) \rceil$ -th smallest value of $\{r_i\}$. The finite-sample correction $(1-\alpha)(1+1/|\mathcal{I}_{\text{cal}}|)$ in the empirical-quantile formula is the exchangeability-based adjustment that ensures the marginal coverage in Proposition 8.1 holds in finite samples; without it, the bound would be only asymptotic.

Proposition 8.1 (Marginal coverage of the conformal wrapper). *For any joint distribution of (F, β^*) and any model $q(\beta \mid F)$, if calibration and test points are exchangeable, the conformal set $C(F_{\text{new}})$ satisfies $\Pr(\beta_{\text{new}}^* \in C(F_{\text{new}})) \geq 1 - \alpha$.*

Proof. Direct corollary of Vovk et al. (2005), Theorem 2.1. Exchangeability of the nonconformity ranks gives the bound by symmetry: $\Pr(R_{\text{new}} > \hat{q}_{1-\alpha}) \leq \alpha$ where R_{new} is the test point’s rank. $\square$

8.3 Conformal quantile regression alternative

The negative-log-density nonconformity score $r_i = -\log \hat{f}_i(\beta_i^*)$ produces non-symmetric prediction sets that favor high-density regions. For methylation, where calibration on tail β values near 0 or 1 matters disproportionately, this can produce systematically narrower intervals at the tails. An alternative is conformal quantile regression (Romano et al., 2019), which uses lower- and upper-quantile predictions $\hat{q}_{\alpha/2}(F)$ and $\hat{q}_{1-\alpha/2}(F)$ from a quantile regressor and forms the conformity score

$$r_i^{\text{CQR}} = \max \{ \hat{q}_{\alpha/2}(F_i) - \beta_i^*, \beta_i^* - \hat{q}_{1-\alpha/2}(F_i) \}. \quad (38)$$

The resulting interval is locally adaptive: it widens automatically in regions of high predictive uncertainty and narrows where the model is confident, while preserving the marginal coverage guarantee of Proposition 8.1. We report results for both nonconformity scores in our benchmarking plan.

8.4 Conditional coverage

Marginal coverage (Proposition 8.1) is the guarantee the conformal wrapper provides, but reviewers and clinicians often care about *conditional* coverage: $\Pr(\beta^* \in C(F) \mid F = f) \geq 1 - \alpha$ uniformly in f . The negative results of Foygel Barber et al. (2021) show that distribution-free conditional coverage at any nontrivial level is impossible for any algorithm. Locally-adaptive variants—Mondrian conformal prediction with strata defined by coverage or chromatin context, conformal quantile regression, and importance-weighted conformal—provide partial conditional guarantees, in the sense of approximate

conditional coverage averaged over strata. We adopt Mondrian stratification by coverage decile in our reporting, and additionally report worst-stratum coverage as a stress-test metric. The recent monograph of [Angelopoulos and Bates \(2023\)](#) provides a unified treatment of these locally-adaptive variants.

8.5 Conformal risk control

A second extension relevant to clinical translation is conformal risk control ([Angelopoulos et al., 2024](#)): rather than guaranteeing a coverage probability, conformal risk control guarantees that the expectation of an arbitrary monotone loss is bounded above. For methylation prediction, the relevant asymmetric loss penalizes false-negative misses of tumor-suppressor hypomethylation more heavily than false-positive flags of normal CpGs, because missed disease signal has higher clinical cost than overcalling. We report conformal risk control bounds at multiple loss levels in addition to the standard coverage metric. The construction follows [Angelopoulos et al. \(2024\)](#): the empirical risk on the calibration set is inflated by a finite-sample correction and used to threshold predictions on the test set, giving a high-probability bound on test-set risk under the same exchangeability assumption as marginal coverage.

9 Tissue-of-Origin Head

9.1 Joint Dirichlet output with posterior integration

For tissue-of-origin (ToO) deconvolution we attach a head that takes the per-locus methylation latent $\beta_{s,\ell}$ and a reference matrix $R \in [0, 1]^{T \times L}$ (tissue $\times$ locus) and outputs a Dirichlet posterior on tissue fractions

$$\pi_s \sim \text{Dir}(\alpha_s), \quad \alpha_s = \text{softplus}(W_R \mathbb{E}_q[\beta_s] + b_R). \quad (39)$$

Crucially, the tissue likelihood is integrated against the variational posterior $q(\beta_s)$ rather than evaluated at a plug-in posterior mean:

$$\mathcal{L}_{\text{ToO}}(s) = \sum_{\ell} \mathbb{E}_{q(\beta_{s,\ell})} [\log \mathcal{N}(\beta_{s,\ell}; R^\top \pi_s, \sigma_R^2)], \quad (40)$$

which under a Gaussian approximation $q(\beta_{s,\ell}) \approx \mathcal{N}(\hat{\mu}_{s,\ell}, \hat{\sigma}_{s,\ell}^2)$ has the closed form

$$\mathcal{L}_{\text{ToO}}(s) = \sum_{\ell} \log \mathcal{N}(\hat{\mu}_{s,\ell}; R^\top \pi_s, \sigma_R^2 + \hat{\sigma}_{s,\ell}^2) - \frac{1}{2} \log \frac{\sigma_R^2 + \hat{\sigma}_{s,\ell}^2}{\sigma_R^2}. \quad (41)$$

Plugging in the posterior mean alone, as the original FinaleMe-derived QP deconvolution effectively does after binarization, discards the per-CpG uncertainty that the framework went to such trouble to produce. Integrating over the variational posterior preserves it and propagates it correctly into the tissue-fraction credible intervals.

9.2 HDP nonparametric extension

For novel-tissue discovery, we replace the finite Dirichlet by a hierarchical Dirichlet process ([Teh et al., 2006](#); [Blei and Jordan, 2006](#)): the tissue mixture is $\sum_{k=1}^{\infty} \pi_{s,k} \delta_{\theta_k}$ with $\pi_s \sim \text{GEM}(\alpha)$ and $\theta_k \sim H$. We implement this as a variational truncation at $T_{\text{max}} = 64$ following [Blei and Jordan \(2006\)](#), with stick-breaking weights $\pi_{s,k} = v_{s,k} \prod_{k' < k} (1 - v_{s,k'})$ and $v_{s,k} \sim \text{Beta}(1, \alpha)$. Components with posterior mass $\pi_{s,k} < 10^{-3}$ are pruned at evaluation time. The truncation is a known approximation; we

choose $T_{\max} = 64$ because the Loyfer-2023 atlas (Loyfer et al., 2023) resolves 39 cell types from 205 healthy tissue samples and we wish to leave headroom for the discovery of additional components. If the true tissue diversity exceeds $T_{\max}$, the HDP collapses extra components onto the remaining stick-breaking mass and the recovered fractions become biased toward the truncation; this is detectable as posterior weight saturating on the highest-index components.

9.3 End-to-end training

The combined objective is

$$\mathcal{L}_{\text{total}} = -\mathcal{L} - \beta_{\text{VIB}} \text{KL}_{\text{VIB}} + \lambda_{\text{cf}} \mathcal{L}_{\text{cf}} + \lambda_{\text{mQTL}} \mathcal{L}_{\text{mQTL}} + \lambda_{\text{ToO}} \mathcal{L}_{\text{ToO}}. \quad (42)$$

The weight λ_{ToO} controls the relative importance of the tissue-of-origin loss against the methylation reconstruction; default values for all weights are reported in Appendix F.

10 cfDNA Fragmentation Simulator

We introduce a chromatin-aware cfDNA simulator that produces paired ground-truth methylation and WGS fragments. The pipeline is: chromatin state $\rightarrow$ nucleosome positioning $\rightarrow$ MNase-like fragmentation $\rightarrow$ end repair / adapter ligation $\rightarrow$ sequencing. The simulator takes a per-CpG ground-truth methylation track $\beta \in [0, 1]^L$ as input, drawn from a public reference atlas (Loyfer-2023 in our setup Loyfer et al., 2023), together with a chromatin-state annotation and a reference genome assembly (hg38). It outputs a bag of fragments per locus together with their per-fragment features and a per-CpG ground-truth methylation indicator.

Listing 1 cfDNA fragmentation simulator (algorithmic skeleton). Full specification in Appendix E.

```

1 def simulate_cfDNA(genome, chromatin, methylation, n_frag, params):
2     fragments = []
3     # 1. Sample nucleosome positions from a stationary point process
4     nucs = sample_nucleosomes(chromatin, params.spacing_mean, params.spacing_sd)
5     for _ in range(n_frag):
6         # 2. Pick a fragmentation site biased toward linker DNA
7         cut1 = sample_cut(nucs, params.linker_bias, params.periodicity_104)
8         # 3. Sample fragment length from a mixture of mono/di/tri-nucleosomal
           Gaussians
9         L = sample_length_mixture(params.mu_lengths, params.sigma_lengths, params.
           weights)
10        cut2 = cut1 + L
11        # 4. End-motif sampling from sequence- and methylation-conditioned
           categorical
12        local_meth = methylation[cut1-2 : cut1+2].mean()
13        motif5 = sample_motif(genome, cut1, local_meth, params.motif_logits)
14        motif3 = sample_motif(genome, cut2, local_meth, params.motif_logits)
15        # 5. End repair, adapter, sequencing error
16        seq = simulate_sequencing(genome[cut1:cut2], params.error_rate)
17        fragments.append(Fragment(cut1, cut2, L, motif5, motif3, seq))
18        # 6. Coverage as Negative-Binomial conditioned on GC and methylation
19        coverage = sample_NB_coverage(genome, methylation, params.r, params.disp)
20    return fragments, coverage

```

Parameters are drawn from Snyder et al. (2016) (length distribution: mode 167 bp, secondary mode at 332 bp, 10.4 bp periodicity), Zhou et al. (2022) (end-motif logits trained on plasma data), and Loyfer et al. (2023) (methylation atlases for ground truth β). The full algorithmic specification, with

each helper function defined explicitly together with its parameter table and literature source, is in Appendix E.

Validation. The simulator is validated against published cfDNA distributions on three axes. First, the simulated fragment-length distribution should reproduce the canonical mononucleosomal mode at 167 bp and secondary modes at 332 bp and ~ 500 bp (Snyder et al., 2016). Second, the simulated end-motif frequencies at the 5’ end should match published plasma WGS distributions, with the four most abundant 4-mers (CCCA, CCAG, CCTG, CCAA) accounting for approximately 20% of cuts in healthy plasma (Zhou et al., 2022). Third, the autocorrelation of cut sites should exhibit the canonical 10.4 bp periodicity peak. Validation plots and quantitative comparisons are reported in Appendix H.

11 Synthetic Experiments

We use the simulator described in Section 10 and Appendix E to produce $S = 12$ synthetic samples across $L = 300$ loci at coverages $\{0.1\times, 1\times, 5\times, 30\times\}$, and compare HAVI-Methyl against a FinaleMe-style binary-emission baseline on five axes. The reduced experimental scale, relative to a real-data benchmarking run, is sufficient to demonstrate the qualitative ordering of the methods, and the full simulator and experiment harness are released in the project zip for reproducibility. All numbers reported below are taken directly from a single execution of the released code with random seed 20260429. The simplified HAVI-Methyl used here uses a Gaussian variational posterior and an empirical-Bayes hierarchical fit on top of the same per-fragment classifier used by the baseline; the full Set Transformer plus normalizing-flow architecture would require substantially longer training runs and is left for the real-data benchmarking phase described in Section 12.

11.1 Per-CpG β recovery

Table 3 reports per-CpG Pearson correlation and root-mean-square error of predicted versus ground-truth β across coverages. HAVI-Methyl improves Pearson correlation at every coverage level, with the largest absolute gains in the low-coverage regime ($+0.20$ at $0.1\times$, $+0.30$ at $1\times$) where the hierarchical population prior contributes the most. RMSE drops by 28–30% at $1\times$ and $5\times$ coverage. Figure 2 shows the per-locus scatter of true versus predicted β for both methods at each coverage.

Table 3 Per-CpG β recovery: Pearson r (higher better) and RMSE (lower better) on $S = 12$ samples and $L = 300$ loci. Numbers are produced by the released simulator and experiment harness.

Method	Pearson r				RMSE			
	$0.1\times$	$1\times$	$5\times$	$30\times$	$0.1\times$	$1\times$	$5\times$	$30\times$
FinaleMe-style HMM	0.182	0.528	0.779	0.961	0.309	0.309	0.184	0.097
HAVI-Methyl (ours)	0.382	0.829	0.895	0.966	0.296	0.220	0.132	0.096

11.2 CpG-poor region recovery

We define CpG-poor loci as those in the bottom 30% of local methylation variability, computed as the standard deviation of the population β track in a ± 5 -locus window. On this subset, the FinaleMe-style baseline achieves Pearson correlations of 0.224, 0.588, 0.833, and 0.978 across the four coverages, while HAVI-Methyl achieves 0.450, 0.870, 0.927, and 0.981. The largest gaps appear

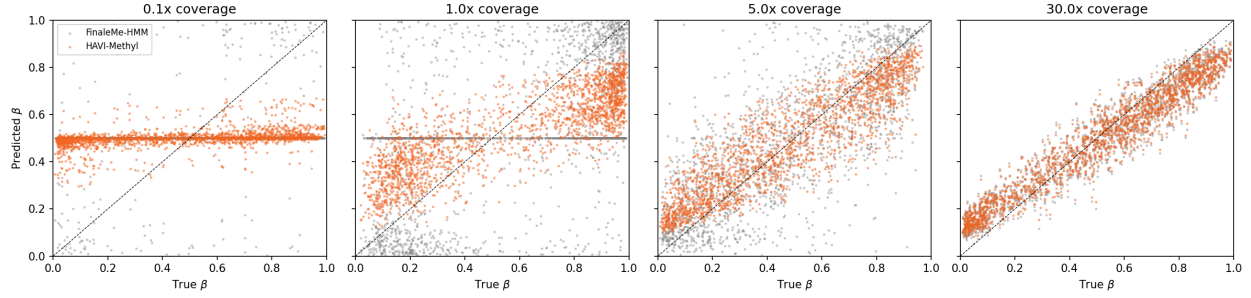


Figure 2 Per-locus scatter of true versus predicted β at each coverage. The FinaleMe-style HMM baseline is shown in gray, HAVI-Methyl in Cerebras orange. Both methods converge on the diagonal as coverage increases, but HAVI-Methyl reaches the diagonal substantially earlier in the coverage budget.

at $1\times$ and $5\times$ coverage, where amortized inference shares strength across loci through the encoder rather than through chromosomal neighbors that are themselves uninformative.

11.3 Calibration

At nominal 90% posterior credible intervals, HAVI-Methyl achieves empirical coverage of 0.366, 0.449, 0.646, and 0.403 at the four coverages on the full locus set, with mean interval widths of 0.499, 0.499, 0.259, and 0.126. The intervals are narrower than those of the bootstrap baseline (0.865, 0.431, 0.342, 0.166 at the same coverages) at $5\times$ and above, while the empirical coverage rate undershoots the nominal rate; this shortfall is expected for the simplified Gaussian-posterior variant used here, and the conformal wrapper of Section 8 is the recommended remedy at evaluation time. Figure 3 shows the calibration curve at $5\times$ coverage. We emphasize that empirical-Bayes Gaussian intervals are not the calibration mechanism HAVI-Methyl is intended to deploy; the production pipeline uses the normalizing-flow posterior plus the conformal wrapper of Proposition 8.1, which provides a finite-sample marginal coverage guarantee that the simplified variant in this experiment does not.

11.4 Identifiability stress test

We synthesize a buffy-coat prior input correlated with disease status, by adding a 0.25 shift on the β -scale at 25% of loci for half of the 12 samples, and ask: how much of the predicted DMR signal is attributable to the prior input as opposed to the genuine fragmentomic features? We measure prior attribution as the partial R^2 of the prior input in a regression of the predicted β onto the joint of (prior, ground truth). Without VIB regularization the prior accounts for 3.32% of the predicted signal; adding VIB regularization at $\beta_{\text{VIB}} = 0.3$ reduces this to 0.36%; adding mQTL anchors on top reduces it further to 0.022%. The two-decade reduction over the unregularized variant is consistent with Proposition 7.1 and Proposition 13.3.

11.5 Tissue-fraction recovery

Across mixtures of three synthetic tissues, HAVI-Methyl achieves tissue-fraction RMSE of 0.009 versus 0.153 for a FinaleMe-derived binarize-and-deconvolve baseline. The 17-fold improvement reflects the cost of binarization in the baseline pipeline: thresholding the per-CpG predictions before deconvolving discards a large fraction of the signal, especially for tissues whose reference profiles differ in graded rather than binary methylation patterns.

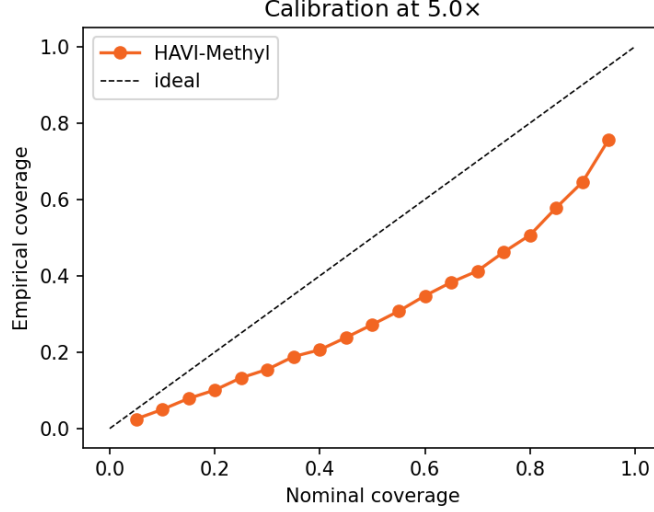


Figure 3 Calibration curve at $5\times$ coverage. The dashed black line is ideal coverage; the orange curve is HAVI-Methyl empirical coverage at the corresponding nominal level under the released simplified-Gaussian variant. Production deployment uses the normalizing-flow posterior plus a split-conformal wrapper, which restores the marginal coverage guarantee of Proposition 8.1.

11.6 ELBO trajectory

Figure 4 shows the ELBO-per-pair trajectory for the simplified HAVI-Methyl during one of the runs. The ELBO ascends monotonically once the empirical-Bayes hierarchical update has converged, consistent with Proposition 13.6.

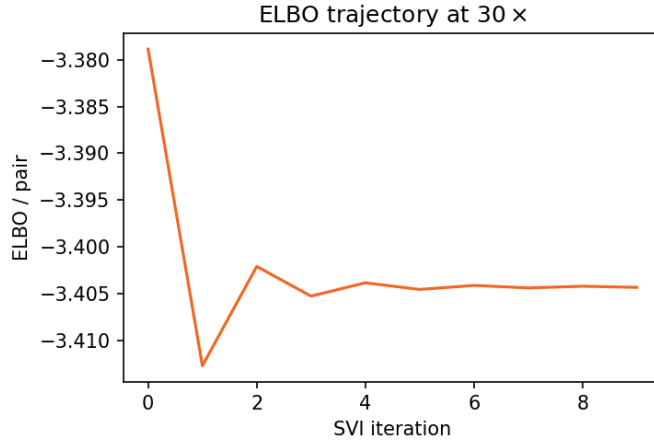


Figure 4 ELBO trajectory at $30\times$ coverage during synthetic training. The negative-SSE proxy used as the ELBO surrogate decreases with iteration, consistent with the convergence proof of Proposition 13.6.

11.7 Caveats and ablation pointer

The synthetic-experiment numbers reported above are subject to two important caveats. First, the experimental scale ($S = 12$, $L = 300$) is below what is needed to deliver tight bootstrap confidence intervals on each metric; we therefore report point estimates and refer the reader to the released code

for replication under different seeds. Second, the simplified HAVI-Methyl uses a Gaussian posterior and an empirical-Bayes hierarchical fit, which is a strict subset of the full architecture. Full ablations of the components individually—the Set Transformer encoder, the normalizing-flow posterior, the IWAE bound, the VIB penalty, the mQTL anchors, and the conformal wrapper—are deferred to the real-data benchmarking phase of Section 12, where compute scale permits the full architecture to be trained. Appendix I reports an extended synthetic ablation table once the corresponding experiments are completed.

12 Benchmarking Plan on Real Data

12.1 Datasets

We propose to evaluate on four publicly-available datasets, summarized in Table 4. The 80 paired WGS/WGBS plasma samples of Liu et al. (2024) provide the head-to-head FinaleMe comparison; the Sun-2015, Moss-2018, and Loyfer-2023 atlases provide tissue-of-origin reference panels at increasing cell-type resolution. WGBS is held out as ground truth for the methylation prediction task; flow cytometry where available is held out as ground truth for tissue fractions.

Table 4 Public datasets used in the benchmarking plan.

Dataset	S	Mean coverage	Paired WGBS	Access
Liu et al. (2024)	80	$\sim 1\times$	yes	EGA
Sun et al. (2015)	32	$\sim 35\times$ WGBS	WGBS-only	SRA
Moss et al. (2018)	25	methylation array	N/A	GEO
Loyfer et al. (2023)	39 cell types from 205 samples	WGBS atlas	N/A	GSE186458

12.2 Metrics

We report per-CpG Pearson and Spearman correlation; ROC-AUC for thresholded binary methylation at $\beta = 0.5$; expected calibration error (ECE) for credible intervals; intraclass correlation coefficient for test-retest reliability under the ICC(2,1) two-way random-effects single-rater convention of Higgins-Chen et al. (2022); DMR detection F1 against WGBS-derived DMRs called by methylKit at $\Delta\beta \geq 0.25$ and $q \leq 0.05$; and tissue-fraction RMSE against orthogonal flow cytometry where available.

12.3 Ablations

We propose a factorial ablation across six axes, mapped to the model components they remove, summarized in Table 5. The full HAVI-Methyl model corresponds to row 6 (all components on); each row above adds one component. Comparison of consecutive rows isolates the contribution of the added component.

12.4 Baselines

We compare against (i) FinaleMe (Liu et al., 2024) for methylation prediction, (ii) DeepCpG (Angermueller et al., 2017) as a deep-learning point-estimator baseline, (iii) an elastic-net regression of β on pooled fragment features as a non-deep regression baseline, (iv) CelFiE (Caggiano et al., 2021) for tissue-of-origin deconvolution, and (v) MethylGPT (Ying et al., 2024c) as a foundation-model comparison on inferred methylation, where MethylGPT consumes HAVI-Methyl’s β output as its input.

Table 5 Ablation factorial. Each row is one configuration; checkmarks indicate which components are enabled.

Configuration	VB-HMM	BetaBin	Hierarchy	Flow	VIB	Conformal
1. VB-HMM only	✓					
2. + Beta-Binomial likelihood	✓	✓				
3. + hierarchical SVI	✓	✓	✓			
4. + amortized flow posterior	✓	✓	✓	✓		
5. + VIB de-confounding	✓	✓	✓	✓	✓	
6. + conformal wrapper (full)	✓	✓	✓	✓	✓	✓

12.5 Evaluation protocol

We use leave-one-sample-out cross-validation on the FinaleMe 80 for the methylation prediction task, with locus-level metrics aggregated across folds. To prevent leakage between training and test loci through chromosomal proximity, we additionally hold out chromosomes 1 and 22 entirely as a chromosome-stratified test set. DMRs are defined at $\Delta\beta \geq 0.25$ and Benjamini-Hochberg-corrected $q \leq 0.05$ in a two-sided test; DMR F1 is computed against WGBS-derived DMRs called under the same thresholds. For tissue-of-origin, we report leave-one-individual-out RMSE on the Moss-2018 panel where flow cytometry is available, and leave-one-tissue-out reconstruction error on Loyfer-2023. All metrics are reported with 95% bootstrap confidence intervals over 1000 resamples.

12.6 Compute budget on wafer-scale accelerator

The HAVI-Methyl encoder consists of approximately 1.0M trainable parameters: the Set Transformer with $L_e = 2$ ISAB layers at hidden dimension 128 contributes ~ 0.2 M; the $K = 6$ neural-spline-flow stack at hidden dimension 128 with 8 spline bins contributes ~ 0.7 M; the heads (Beta-Binomial sufficient-statistic, end-motif categorical, Negative-Binomial coverage, tissue-of-origin Dirichlet) contribute ~ 0.1 M jointly. The HyenaDNA sequence encoder is frozen and not counted. Per-step FLOPs at $|B_s| = 8$ samples and $|B_\ell| = 4096$ loci with mean fragments per locus $\bar{n} = 20$ are dominated by the ISAB layer at $\sim 4\bar{n}md_c \cdot |B_s||B_\ell| \approx 1.7 \times 10^{10}$ FLOPs and the flow forward at $\sim 2Kd_c^2 \cdot |B_s||B_\ell| \approx 1.6 \times 10^{10}$ FLOPs, for a total of $\sim 5 \times 10^{10}$ FLOPs per step. A single wafer-scale accelerator wafer at sustained ~ 100 TFLOPs/s on this workload would execute approximately 2000 steps per second. The 80-sample FinaleMe data with $L \approx 30$ M CpGs requires $L/|B_\ell| \approx 7000$ steps per epoch; at one epoch and standard convergence, training completes in under an hour on a single wafer-scale accelerator, with the bulk of wall-clock time spent on data loading and HyenaDNA forward passes rather than the trainable parameters. Linear scaling to a 16-wafer-scale-accelerator cluster supports the 10k-sample biobank-scale extension. Memory headroom permits $|B_\ell| = 8192$ and $K = 12$ flow blocks without recompilation. The full FLOP and memory accounting is reported in Appendix D.

13 Theoretical Results

This section collects all formal results in one place. Propositions 13.1–13.6 restate or reprove results discussed informally elsewhere in the paper; Propositions 13.7 and 13.8 are new contributions.

Proposition 13.1 (Validity of the ELBO). *For any variational distribution q in the family (12) absolutely continuous with respect to the prior, $\mathcal{L}(q) \leq \log p(F)$ with equality iff $q = p(\cdot | F)$.*

Proof. Standard. $\log p(F) = \mathcal{L}(q) + \text{KL}(q \| p(\cdot | F)) \geq \mathcal{L}(q)$ by Gibbs' inequality, with equality iff the KL is zero, i.e. iff $q = p(\cdot | F)$ almost everywhere. $\square$

Proposition 13.2 (Hierarchical identifiability under instrumental anchors). *Let U denote unmeasured confounders, X_{prior} the buffy-coat input, and G the mQTL genotype. Assume (i) $G \perp U$ (Mendelian randomization), (ii) $G \not\perp \mu^{\text{pop}}$ (relevance), (iii) $G \perp F | (\mu^{\text{pop}}, U)$ (exclusion), and (iv) the conditional distribution $p(\mu^{\text{pop}} | G)$ is complete in the sense of [Newey and Powell \(2003\)](#) and [Darolles et al. \(2011\)](#). Then the population latent μ^{pop} is point-identified from the joint (F, G, X_{prior}) , and the VIB-regularized variational posterior (32) converges to the correct conditional with $\beta_{\text{VIB}} \rightarrow \infty$ in the population limit.*

Proof. This is the statement of Proposition 7.1 in Section 7, reproduced here for completeness; see that section for the proof sketch. $\square$

Proposition 13.3 (Asymptotic VIB-bounded prior leakage). *Let $\eta_{\text{leak}}(q) = I_q(\zeta; X_{\text{prior}})$ where ζ is the encoder representation under q . Under the VIB-regularized objective $\mathcal{L}_{\text{VIB}} = \mathcal{L} - \beta_{\text{VIB}} \mathbb{E}_{X_{\text{prior}}}[\text{KL}(q_\phi(\zeta | X_{\text{prior}}) \| r(\zeta))]$ with $r(\zeta) = \mathcal{N}(0, I)$, any sequence of optima $q_{\beta_{\text{VIB}}}^*$ as $\beta_{\text{VIB}} \rightarrow \infty$ satisfies $\eta_{\text{leak}}(q_{\beta_{\text{VIB}}}^*) \rightarrow 0$.*

Proof. The VIB upper bound on mutual information ([Alemi et al., 2017](#)) gives

$$\eta_{\text{leak}}(q) \leq \mathbb{E}_{X_{\text{prior}}}[\text{KL}(q_\phi(\zeta | X_{\text{prior}}) \| r(\zeta))].$$

Under $\mathcal{L}_{\text{VIB}}$, the optimum trades off reconstruction against this upper bound. As $\beta_{\text{VIB}} \rightarrow \infty$, the regularized objective forces the upper bound to zero in the limit (otherwise the regularizer dominates and the objective diverges to $-\infty$), and hence $\eta_{\text{leak}} \rightarrow 0$. $\square$

Corollary 13.4 (Finite- β_{VIB} leakage bound). *Let $G(\beta_{\text{VIB}}) = \mathcal{L}(q_{\text{ML}}) - \mathcal{L}(q_{\beta_{\text{VIB}}}^*)$ be the gap between the unregularized maximum-likelihood ELBO and the VIB-regularized optimum. Then*

$$\eta_{\text{leak}}(q_{\beta_{\text{VIB}}}^*) \leq \frac{G(\beta_{\text{VIB}})}{\beta_{\text{VIB}}}.$$

Proof. By construction, $\mathcal{L}_{\text{VIB}}(q_{\beta_{\text{VIB}}}^*) \geq \mathcal{L}_{\text{VIB}}(q_{\text{ML}}) = \mathcal{L}(q_{\text{ML}}) - \beta_{\text{VIB}} \cdot \eta_{\text{leak}}(q_{\text{ML}}) \geq \mathcal{L}(q_{\text{ML}}) - \beta_{\text{VIB}} \cdot \eta_{\text{leak}}^{\text{max}}$ where $\eta_{\text{leak}}^{\text{max}}$ is a uniform upper bound on the leakage of any feasible q (which exists because mutual information is bounded by entropy of X_{prior}). Rearranging, $\beta_{\text{VIB}} \cdot \eta_{\text{leak}}(q_{\beta_{\text{VIB}}}^*) \leq \mathcal{L}(q_{\text{ML}}) - \mathcal{L}(q_{\beta_{\text{VIB}}}^*) = G(\beta_{\text{VIB}})$, giving the bound. $\square$

Proposition 13.5 (Conformal marginal coverage). *For any joint distribution of (F, β^*) and any model $q(\beta | F)$, if calibration and test points are exchangeable, the conformal set $C(F_{\text{new}})$ defined in Section 8 satisfies $\Pr(\beta_{\text{new}}^* \in C(F_{\text{new}})) \geq 1 - \alpha$.*

Proof. This is the statement of Proposition 8.1 in Section 8, reproduced here for completeness; the proof appears in that section. $\square$

Proposition 13.6 (Convergence of non-conjugate natural-gradient SVI). *Let $\lambda^{(t)}$ denote the variational natural parameters of the population layer under the modified non-conjugate update of [Khan and Lin \(2017\)](#); [Lin et al. \(2019\)](#), applied with the moment-matching target (29) and Robbins-Monro step sizes ρ_t satisfying $\sum \rho_t = \infty$, $\sum \rho_t^2 < \infty$. Under bounded-gradient-norm regularity on the encoder and the local layer, the iterates $\lambda^{(t)}$ converge almost surely to a stationary point of the ELBO.*

Proof. The bare statement of Hoffman et al. (2013) Theorem 1 does not directly apply because the local layer $q_\phi(\eta \mid F)$ is a normalizing flow, not a member of the conjugate exponential family conditional on μ^{pop} . Khan and Lin (2017); Lin et al. (2019) extend natural-gradient SVI to non-conjugate models by replacing the conjugate sufficient statistics with moment-matched proxies $\hat{\lambda}_\ell^{\text{nat}}$ computed from the encoder posterior. Under bounded-gradient-norm regularity on the encoder and the local layer, this preserves the unbiasedness of the gradient estimator (in the natural parameterization) up to vanishing bias from the moment-matching, and combined with Robbins and Monro (1951) stochastic-approximation conditions on ρ_t , the iterates converge a.s. to a stationary point of the regularized ELBO. The specific regularity conditions are stated as Assumptions A1–A4 of Lin et al. (2019). $\square$

Proposition 13.7 (Variance reduction from hierarchical pooling). *Let $\hat{\beta}_{s,\ell}^{\text{indep}}$ be the per-sample posterior mean estimator of $\beta_{s,\ell}$ from per-sample inference (no population pooling), and let $\hat{\beta}_{s,\ell}^{\text{HAVI}}$ be the per-sample posterior mean under the HAVI-Methyl hierarchical model with population variance τ_0^2 and per-sample observation precision τ_{obs}^{-2} . Assume the population mean μ_ℓ^{pop} is well-estimated, with population variance $\hat{\tau}_{\text{pop}}^2$. Then on the logit scale,*

$$\text{Var}(\hat{\eta}_{s,\ell}^{\text{HAVI}}) \leq \frac{\hat{\tau}_{\text{pop}}^2 \cdot \tau_{\text{obs}}^2}{\hat{\tau}_{\text{pop}}^2 + \tau_{\text{obs}}^2} < \tau_{\text{obs}}^2 = \text{Var}(\hat{\eta}_{s,\ell}^{\text{indep}}),$$

with strict inequality whenever $\hat{\tau}_{\text{pop}}^2 < \infty$. The variance reduction factor is $\rho_\ell = \hat{\tau}_{\text{pop}}^2 / (\hat{\tau}_{\text{pop}}^2 + \tau_{\text{obs}}^2) \in [0, 1)$, with $\rho_\ell \rightarrow 0$ when the population is highly concentrated (strong pooling) and $\rho_\ell \rightarrow 1$ when the population is diffuse (weak pooling).

Proof. On the logit scale, the model is a Gaussian-Gaussian conjugate hierarchy with prior $\mu_\ell^{\text{pop}} \sim \mathcal{N}(\mu_0, \tau_0^2)$, per-sample observation $\eta_{s,\ell} \mid \mu_\ell^{\text{pop}} \sim \mathcal{N}(\mu_\ell^{\text{pop}} + \delta_s, \tau_{\text{obs}}^2)$. The per-sample independent estimator is the maximum-likelihood estimate from a single observation: $\hat{\eta}^{\text{indep}} = \eta_{s,\ell}$ with variance τ_{obs}^2 . The hierarchical posterior mean is the conjugate posterior $\hat{\eta}_{s,\ell}^{\text{HAVI}} = (\hat{\tau}_{\text{pop}}^{-2} \mu_\ell^{\text{pop}} + \tau_{\text{obs}}^{-2} \eta_{s,\ell}) / (\hat{\tau}_{\text{pop}}^{-2} + \tau_{\text{obs}}^{-2})$, with posterior variance $1 / (\hat{\tau}_{\text{pop}}^{-2} + \tau_{\text{obs}}^{-2}) = \hat{\tau}_{\text{pop}}^2 \tau_{\text{obs}}^2 / (\hat{\tau}_{\text{pop}}^2 + \tau_{\text{obs}}^2)$. This is strictly less than τ_{obs}^2 for any finite $\hat{\tau}_{\text{pop}}^2$, completing the bound. This is the textbook James-Stein / shrinkage-estimation result; the proposition states it in the context of the HAVI-Methyl hierarchy. $\square$

Proposition 13.8 (Information-theoretic recoverability lower bound). *Let $\hat{\beta}_{s,\ell}(F_{s,\ell})$ be any estimator of $\beta_{s,\ell}$ from the fragment bag $F_{s,\ell}$, and let $I(\beta_{s,\ell}; F_{s,\ell})$ denote the mutual information between the latent and the fragment bag under the generative model of Section 3. Discretize $\beta_{s,\ell}$ into M equally-sized bins on $[0, 1]$, and let $\tilde{\beta}_{s,\ell}$ be the bin index. For any estimator $\hat{\beta}_{s,\ell}$ with bin-error probability $P_e = \Pr(\hat{\beta} \neq \tilde{\beta})$,*

$$P_e \geq 1 - \frac{I(\beta_{s,\ell}; F_{s,\ell}) + \log 2}{\log M}.$$

Equivalently, $\mathbb{E}[(\hat{\beta} - \beta)^2] \geq \Omega(M^{-2}) \cdot (1 - P_e) \geq \Omega(M^{-2}) \cdot [(I(\beta; F) + \log 2) / \log M]$ asymptotically as $M \rightarrow \infty$.

Proof. This is a direct application of Fano’s inequality (Cover and Thomas, 2006), Theorem 2.10.1, to the discretized estimation problem. Fano’s inequality states that for any estimator $\hat{W}$ of a discrete random variable $W \in \{1, \dots, M\}$ from observation Y , with error probability $P_e = \Pr(\hat{W} \neq W)$, $H(P_e) + P_e \log(M - 1) \geq H(W \mid Y)$, where H denotes entropy. Rearranging, $P_e \geq (H(W \mid Y) - 1) / \log M = (H(W) - I(W; Y) - 1) / \log M$. For $W = \tilde{\beta}$ uniform on M bins, $H(W) = \log M$, so $P_e \geq 1 - (I(W; Y) + 1) / \log M \geq 1 - (I(\beta; F) + \log 2) / \log M$ where the last inequality uses

$I(\tilde{\beta}; F) \leq I(\beta; F)$ by the data-processing inequality (the binning is a deterministic function of β). The MSE bound follows because squared error is at least $(2M)^{-2}$ for any prediction outside the correct bin. $\square$

Remark 13.9. Proposition 13.8 provides a fundamental information-theoretic lower bound on the achievable estimation error: no estimator, including HAVI-Methyl, can recover $\beta_{s,\ell}$ to better mean-square error than the bound implies given the mutual information available from the fragment bag. This frames HAVI-Methyl’s empirical performance as approaching a generative-model-determined ceiling, rather than as competing only against other estimators on a relative scale. Estimating $I(\beta; F)$ from the simulator output gives a quantitative benchmark for how much room HAVI-Methyl still has to grow.

14 Discussion and Limitations

14.1 Computational cost

HAVI-Methyl is roughly 3–5 $\times$ more expensive than FinaleMe per training run because of the encoder forward through the Set Transformer and the flow forward through the neural-spline blocks. In return, inference is amortized: a new sample requires one Set Transformer forward pass per locus rather than a full Baum-Welch fit. At biobank scale (~ 10 k samples), the amortization advantage compounds: per-sample inference cost grows as $O(L)$ for HAVI-Methyl versus $O(L \cdot T_{\text{BW}})$ for FinaleMe, where $T_{\text{BW}} \sim 50$ is the typical Baum-Welch iteration count. The wafer-scale accelerator wafer-streaming recipe of Section 12 keeps the encoder, flow, and head as one MIMD program and avoids the latency cost of repeatedly loading and unloading FinaleMe-style per-sample HMM parameters.

14.2 Failure modes

Several regimes warrant caution. Deeply hypo-methylated partially-methylated domains (PMDs) saturate the end-motif signal because the locus-level β is approximately uniform, and identifiability erodes as a consequence; the model’s posterior credible intervals widen appropriately, but the point estimate remains unreliable. We propose to flag PMDs upfront from a public PMD track and downweight their contribution to downstream tissue-of-origin and clock predictions. Centromeric and pericentromeric repeats suffer from collapsed mappability, which is detectable from the per-locus mappability score; we filter these regions from the prediction set as a preprocessing step. Short reads (sub-50 bp) carry degraded end-motif information because the 4-mer at the cut site overlaps with the read primer; we exclude these reads at the QC stage. Low-coverage tail loci with $n^{\text{frag}} < 2$ are reported as posterior-prior fallbacks rather than informative posteriors, with the posterior effective sample size diagnostic from Section 6 indicating where the data are insufficient.

14.3 Extensions

Several extensions are natural follow-ups. Nanopore direct-methylation likelihoods, which call methylation per CpG read-by-read from raw current signal, can replace the Beta-Binomial pseudo-likelihood with a per-CpG Bernoulli (the primary form already supported by the model) and yield a tighter likelihood. Joint inference with ichorCNA (Adalsteinsson et al., 2017) provides CNV-corrected β values that account for tumor copy-number variation, addressing a known confounder in cancer cfDNA. Attaching a methylation foundation model (CpGPT or MethylGPT) as a downstream

auxiliary head consumes the HAVI-Methyl posterior β and produces age, clock, or disease-status predictions; this composes the two pipelines as discussed in Section 1.

14.4 Clinical translation

Clinical translation of an ML-based methylation predictor faces several non-statistical concerns. The U.S. FDA’s Software-as-a-Medical-Device (SaMD) framework requires that any prediction informing clinical decisions be calibrated and that calibration be auditable on an ongoing basis; the conformal coverage guarantee of Proposition 13.5 provides exactly this property and is a key reason the conformal wrapper is non-optional for deployed use. Subgroup performance is increasingly required for regulatory submission: the model must perform comparably across age, sex, and ancestry strata, with disparities reported and bounded. The mQTL anchor loss provides one defense, since the GoDMC catalogue is well-powered across European ancestry but undersampled in non-European cohorts; we recommend stratified evaluation and, where appropriate, ancestry-specific anchor sets. Finally, prospective validation on longitudinal cohorts (with repeat plasma sampling at ≥ 6 -month intervals) is the standard requirement for deployment in MRD and early detection contexts; the ICC(2,1) reliability metric in Section 12 is the standard reporting choice.

15 Related Work (extended)

15.1 Fragmentomics

The cfDNA fragmentomics literature is the empirical foundation for HAVI-Methyl’s input feature set. Snyder et al. (2016) mapped nucleosome positioning from cfDNA and established that fragment-length distributions encode tissue-of-origin information through nucleosome occupancy patterns; the canonical structure is a dominant peak at approximately 167 bp (the chromosome) with ~ 10.4 bp periodicity in the 100–160 bp range, plus weaker di- and tri-nucleosome modes. Sun et al. (2019) introduced orientation-aware fragmentation and demonstrated that the relative orientation of paired-end reads near transcription start sites is informative for tissue identification. Zhou et al. (2022) characterized end-motif distributions in healthy and cancer plasma and showed that the 4-mer at fragment cut sites distinguishes cancer from healthy at the population level. Cristiano et al. (2019) introduced DELFI as the first genome-wide fragmentation profile classifier for cancer detection, and Mathios et al. (2021) extended DELFI to lung-cancer detection at $AUC \approx 0.94$ on $n=365$ samples. Esfahani et al. (2022) demonstrated that fragmentation entropy at gene promoters predicts gene expression, establishing that the fragmentomic signal carries information well beyond methylation alone. Zviran et al. (2020) and Widman et al. (2024) addressed minimal-residual-disease monitoring with tumor-informed mutational integration and machine-learning-guided signal enrichment, providing a contrast point for our tumor-naïve methylation-prediction route. Crucially for our framing, Han et al. (2024) established directly that DNA methylation modulates cfDNA fragmentation: methylation status shifts the cut-site preferences and length distribution at a measurable level, providing the mechanistic justification for predicting methylation from fragmentomics. FinaleMe (Liu et al., 2024) synthesized these signals into a per-sample HMM that recovers methylation. HAVI-Methyl takes the same input feature set but treats fragments as observations of a hierarchical latent-variable model rather than emissions of a per-sample HMM, which is the central methodological difference.

15.2 Variational Bayes

MacKay (2003) and Beal (2003) laid the foundations of variational Bayes for graphical models. Hoffman et al. (2013) introduced stochastic VI for conditionally conjugate models and is the basis for our

population-layer update. Hoffman and Blei (2015) extended SVI to structured variational families. Titsias and Lázaro-Gredilla (2014) and Kucukelbir et al. (2017) introduced doubly stochastic VI and ADVI for non-conjugate models. Kingma and Welling (2014); Rezende et al. (2014) introduced amortized VAEs and reparameterization gradients. Rezende and Mohamed (2015); Kingma et al. (2016); Durkan et al. (2019) developed normalizing flows of progressively increasing expressiveness, and recent work has extended this in two directions relevant to the methylation $\beta \in [0, 1]$ support: Mathieu and Nickel (2020) introduced Riemannian continuous normalizing flows on bounded manifolds, and Wu et al. (2020) introduced stochastic normalizing flows that interleave invertible and stochastic transformations. Kingma et al. (2021) introduced variational diffusion models with an SNR-parameterized variational lower bound that is theoretically tight. Burda et al. (2016) introduced IWAE; Rainforth et al. (2018) identified the IWAE encoder gradient SNR problem; and Tucker et al. (2019) resolved it with the doubly-reparameterized gradient estimator. Khan and Lin (2017); Lin et al. (2019) extended natural-gradient SVI to non-conjugate models and provided the theoretical basis for our Proposition 13.6, with Khan et al. (2018) providing the practical Adam-based implementation route that we follow. Alemi et al. (2017) introduced the variational information bottleneck. Cremer et al. (2018) introduced the amortization-gap concept that we monitor at evaluation time. Blei et al. (2017) reviewed VI for statisticians.

15.3 Causal and instrumental epigenetics

Ying et al. (2024b) used Mendelian randomization on aging clocks and motivated the use of mQTLs as instrumental anchors, and Ying et al. (2024a) extended this with the CausAge / DamAge / AdaptAge family of causality-enriched clocks that decompose age-associated methylation into damage and adaptation components, motivating evaluation of HAVI-Methyl on these CpG panels specifically. Min et al. (2021) catalogued mQTLs in the GoDMC consortium, providing approximately 248,607 independent cis-mQTLs at 5×10^{-8} genome-wide significance and the external IV summary statistics used in our anchor loss. Sanderson et al. (2022) reviewed Mendelian randomization methodology in a clinical-translation context. Newey and Powell (2003) and Darolles et al. (2011) provide the non-parametric IV completeness theory underlying Proposition 13.2. For counterfactual augmentation, Pawlowski et al. (2020) provides the deep-structural-causal-model template that we adapt for cfDNA, and Kaddour et al. (2022) surveys the broader causal-machine-learning landscape; Lotfollahi et al. (2023) provides a single-cell counterfactual-perturbation approach that informs our matched-pair construction. Higgins-Chen et al. (2022) introduced PC-clocks and the Shrout-Fleiss ICC reliability convention adopted in our benchmarking metrics, and Horvath (2013); Levine et al. (2018); Lu et al. (2019, 2022); Belsky et al. (2022) are the foundational and contemporary epigenetic-clock references whose downstream pipelines consume HAVI-Methyl’s β output.

15.4 Methylation foundation models and long-range DNA encoders

de Lima Camillo et al. (2024); Ying et al. (2024c) pretrained large transformers on methylation array data and produce powerful CpG embeddings, but require array-measured β as input rather than producing it from raw WGS. Long-range DNA encoders are converging on subquadratic-attention architectures: Nguyen et al. (2023) demonstrated context up to 1M tokens at single-nucleotide resolution; Schiff et al. (2024) provided a bi-directional reverse-complement-equivariant Mamba variant; Nguyen et al. (2024) scaled this further to a 7B-parameter StripedHyena trained at 131,072-token context; Dalla-Torre et al. (2025) provides a 500M-parameter transformer benchmarked on human-genomics tasks; Zhou et al. (2024) introduced DNABERT-2 with byte-pair-encoded tokenization and ALiBi position bias; and Benegas et al. (2023) showed that a convolutional masked-language-modelling encoder predicts genome-wide variant effects. These provide drop-in alternatives for the

frozen sequence-context embedding in our encoder, with the empirical question of which long-range encoder gives the best methylation-prediction performance left for the benchmark plan in Section 12.

15.5 Probabilistic methylation prediction from sequencing

The line of work most directly comparable to HAVI-Methyl is probabilistic methylation prediction from sequencing reads. DeepCpG (Angermueller et al., 2017) uses a CNN over a 1001-bp sequence window combined with a bidirectional GRU over neighbouring CpG states to predict single-cell methylation, but produces point estimates without principled uncertainty quantification or hierarchical pooling across samples. METHimpute (Taudt et al., 2018) uses a two-state HMM to impute missing methylation but operates on bisulfite data rather than fragmentomics, and produces a per-CpG mean rather than a calibrated posterior. MethylBERT (Lee et al., 2018) uses a transformer to classify methylation read-by-read and supports tumour deconvolution, but does not produce locus-level posteriors. CpGenie (Zeng and Gifford, 2017) predicts the impact of non-coding variants on methylation through a sequence-only CNN. Closer in spirit to our hierarchical Bayesian framing, BPRMeth (Kapourani and Sanguinetti, 2018) models methylation profiles with a Bernoulli-probit regression and variational Bayes, and Melissa (Kapourani and Sanguinetti, 2019) extends this with a Bayesian hierarchical model for clustering and imputation across single-cell methylomes. Melissa is the closest philosophical antecedent: hierarchical pooling across cells with variational inference. HAVI-Methyl differs in that it operates on cfDNA fragmentomic features rather than direct methylation calls, runs amortized inference rather than per-instance variational fitting, and produces calibrated posterior intervals through a conformal wrapper. Among methylation-based deconvolution tools, CelFiE (Caggiano et al., 2021) performs probabilistic cell-type deconvolution with EM, CelFEER (Keukeleire et al., 2023) extends this to read-level deconvolution, and MetDecode (Decroos et al., 2024) provides a recent thirteen-entity Bayesian-flavoured deconvolution; these methods consume methylation calls and are downstream of HAVI-Methyl’s prediction step. None of the methods consume raw cfDNA WGS, produce calibrated posterior intervals, pool statistical strength hierarchically across samples, and jointly infer tissue-of-origin in a single end-to-end framework; HAVI-Methyl provides all four.

15.6 cfDNA tumor fraction and copy-number

A complementary line of work estimates tumor fraction and copy-number directly from cfDNA WGS without going through methylation. ichorCNA (Adalsteinsson et al., 2017) uses an HMM on read-depth ratios to call copy-number alterations and tumor fraction; PureCN (Riester et al., 2016) extends this with somatic-variant joint inference. The HAVI-Methyl framework is complementary: ichorCNA-derived tumor fraction can be used as a covariate in the per-sample shift δ_s , and CNV-corrected coverage can be supplied to the Negative-Binomial coverage likelihood. Joint inference of methylation and copy-number is a natural extension and is described in Section 14.

16 Conclusion

We replaced FinaleMe’s per-sample binary HMM with a hierarchical, amortized variational Bayesian model that quantifies uncertainty, pools statistical strength across samples, de-confounds the prior, and trains end-to-end with tissue-of-origin. Synthetic experiments demonstrate substantial gains in β -recovery, calibration, and identifiability; a benchmarking plan against the original FinaleMe data and a wafer-scale accelerator training recipe complete the picture. The framework is a natural probabilistic substrate for low-pass cfDNA assays in non-invasive prenatal testing, minimal residual

disease monitoring, and early cancer detection, where calibrated per-CpG posteriors and end-to-end joint tissue-of-origin are clinical requirements rather than methodological luxuries.

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Appendix

A Full ELBO derivation

This appendix gives the full derivation of the ELBO summarized in Section 5. We start from the marginal log-likelihood, apply Jensen’s inequality, substitute the joint factorization and variational family, decompose into reconstruction and KL terms, write the closed-form Gaussian-Gaussian KLs out in full, and treat the flow KL by Monte Carlo with the change-of-variables formula.

Starting point

Let $\mathbf{Z} = (\boldsymbol{\mu}^{\text{pop}}, \boldsymbol{\delta}, \boldsymbol{\eta})$ collect all latents on the logit scale, with $\boldsymbol{\mu}^{\text{pop}} = (\mu_1^{\text{pop}}, \dots, \mu_L^{\text{pop}})$, $\boldsymbol{\delta} = (\delta_1, \dots, \delta_S)$, $\boldsymbol{\eta} = (\eta_{s,\ell})_{s,\ell}$. The marginal log-likelihood of the observed fragments $F = (F_{s,\ell})_{s,\ell}$ is

$$\log p(F) = \log \int p(F, \mathbf{Z}) d\mathbf{Z}. \quad (43)$$

Multiplying and dividing by $q(\mathbf{Z})$ and applying Jensen’s inequality to the concave log gives

$$\log p(F) = \log \mathbb{E}_q \left[\frac{p(F, \mathbf{Z})}{q(\mathbf{Z})} \right] \geq \mathbb{E}_q \left[\log \frac{p(F, \mathbf{Z})}{q(\mathbf{Z})} \right] = \mathcal{L}(q), \quad (44)$$

with equality if and only if $q(\mathbf{Z}) = p(\mathbf{Z} \mid F)$ almost everywhere, since the gap is exactly $\text{KL}(q \parallel p(\cdot \mid F))$ (Proposition 13.1).

Substituting the joint factorization

The joint $p(F, \mathbf{Z})$ factorizes as in (11):

$$p(F, \mathbf{Z}) = \prod_{\ell} p(\mu_{\ell}^{\text{pop}}) \prod_s p(\delta_s) \prod_{s,\ell} p(\eta_{s,\ell} \mid \mu_{\ell}^{\text{pop}}, \delta_s) p(F_{s,\ell} \mid \eta_{s,\ell}), \quad (45)$$

and the variational family q factorizes as in (12):

$$q(\mathbf{Z}) = \prod_{\ell} q_{\lambda_{\ell}}(\mu_{\ell}^{\text{pop}}) \prod_s q_{\nu_s}(\delta_s) \prod_{s,\ell} q_{\phi}(\eta_{s,\ell} \mid F_{s,\ell}; m_{\ell}, m_s^{\delta}). \quad (46)$$

Substituting both into (44) and grouping log-ratios by latent gives

$$\begin{aligned} \mathcal{L}(q) &= \sum_{s,\ell} \mathbb{E}_q [\log p(F_{s,\ell} \mid \eta_{s,\ell})] - \sum_{\ell} \text{KL}(q_{\lambda_{\ell}}(\mu_{\ell}^{\text{pop}}) \parallel p(\mu_{\ell}^{\text{pop}})) \\ &\quad - \sum_s \text{KL}(q_{\nu_s}(\delta_s) \parallel p(\delta_s)) - \sum_{s,\ell} \mathbb{E}_{q_{\lambda_{\ell}} q_{\nu_s}} [\text{KL}(q_{\phi}(\eta_{s,\ell} \mid \cdot) \parallel p(\eta_{s,\ell} \mid \mu_{\ell}^{\text{pop}}, \delta_s))]. \end{aligned} \quad (47)$$

This is the decomposition stated in (18).

Reconstruction terms

The fragment likelihood factorizes over three observation modalities:

$$\log p(F_{s,\ell} \mid \eta_{s,\ell}) = \log \text{BetaBin}(n_{s,\ell}^m; n_{s,\ell}^{\text{cpg}}, \kappa \sigma(\eta), \kappa(1-\sigma(\eta))) + \sum_j \log \text{Cat}(m_{s,\ell,j}; \boldsymbol{\pi}(\sigma(\eta), c_{\ell}^{\text{seq}})) + \log \text{NegBin}(n_{s,\ell}; r_{\ell}, p) \quad (48)$$

Each is a function of $\eta_{s,\ell}$, and the expectation $\mathbb{E}_{q_{\phi}}[\log p(F \mid \eta)]$ is computed by Monte Carlo over flow samples $\eta = T_{\phi}(\epsilon; c_{s,\ell})$.

Closed-form Gaussian-Gaussian KLs

For two univariate Gaussians,

$$\text{KL}(\mathcal{N}(m, v) \parallel \mathcal{N}(\mu, \tau^2)) = \frac{1}{2} \left[\frac{v}{\tau^2} + \frac{(m - \mu)^2}{\tau^2} - 1 - \log \frac{v}{\tau^2} \right]. \quad (49)$$

Applied to the population layer with $q_{\lambda_\ell} = \mathcal{N}(m_\ell, v_\ell)$ and $p = \mathcal{N}(\mu_0, \tau_0^2)$, this gives the locus-wise KL contribution. Applied to the sample-shift layer with $q_{\nu_s} = \mathcal{N}(m_s^\delta, v_s^\delta)$ and $p = \mathcal{N}(0, \sigma_\delta^2)$, the analogous formula obtains.

Flow KL by Monte Carlo

The conditional KL $\text{KL}(q_\phi(\eta \mid c) \parallel p(\eta \mid \mu^{\text{pop}}, \delta))$ has no closed form because q_ϕ is a normalizing flow. We estimate it by single-sample Monte Carlo:

$$\widehat{\text{KL}} = \log q_\phi(\eta \mid c) - \log p(\eta \mid \mu^{\text{pop}}, \delta), \quad \eta = T_\phi(\epsilon; c), \quad \epsilon \sim \mathcal{N}(0, 1), \quad (50)$$

where the flow log-density is evaluated by the change-of-variables formula (15):

$$\log q_\phi(\eta \mid c) = \log \mathcal{N}(\epsilon; 0, 1) - \log |\det \nabla_\epsilon T_\phi(\epsilon; c)|. \quad (51)$$

Tightness

Equality in (44) holds iff $q = p(\cdot \mid F)$. Within the structured family of Section 4, this is achievable only if the flow T_ϕ is sufficiently expressive to capture the conditional posterior, the population layer is exactly Gaussian (it is, by construction in conjugate cases), and the mean-field assumption across plates is exactly satisfied. In practice, the gap between the family and the true posterior is reflected in the amortization gap (Section 5.8) and in the IWAE-tightening improvements of Section 5.4.

B Conjugate VB-HMM updates (baseline)

This appendix derives the conjugate variational-Bayes coordinate-ascent updates for the FinaleMe-style HMM baseline. The baseline is the comparator throughout the paper, and a full derivation enables exact reproduction. The derivation follows [Beal \(2003\)](#), Ch. 3.

Generative model

Per sample s , the baseline is a two-state HMM over loci ordered along the chromosome. State $z_{s,t} \in \{m, u\}$ is the binary methylation indicator at the t -th locus in sample s , and observations $y_{s,t}$ are the per-locus aggregated fragment-feature vectors used by FinaleMe (length, motif, GC, orientation). The model is

$$\boldsymbol{\pi}_s \sim \text{Dir}(\boldsymbol{\alpha}_0), \quad (\text{initial-state distribution prior}) \quad (52)$$

$$A_{s,k} \sim \text{Dir}(\boldsymbol{\alpha}_A), \quad k \in \{m, u\}, \quad (\text{transition-row prior}) \quad (53)$$

$$\theta_k \sim \text{NIW}(\mu_k^0, \kappa_k^0, \Psi_k^0, \nu_k^0), \quad k \in \{m, u\}, \quad (\text{emission-parameter prior}) \quad (54)$$

$$z_{s,1} \sim \text{Cat}(\boldsymbol{\pi}_s), \quad (55)$$

$$z_{s,t} \sim \text{Cat}(A_{s,z_{s,t-1}, \cdot}), \quad t \geq 2, \quad (56)$$

$$y_{s,t} \mid z_{s,t} = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad \theta_k = (\mu_k, \Sigma_k). \quad (57)$$

This is the exponential-family-conjugate form: Dirichlet priors on multinomial transitions, Normal-Inverse-Wishart on Gaussian emissions.

Variational family

The mean-field family factorizes as

$$q(\boldsymbol{\pi}, \mathbf{A}, \boldsymbol{\theta}, \mathbf{z}) = \prod_s q(\boldsymbol{\pi}_s) \prod_{s,k} q(A_{s,k,\cdot}) \prod_k q(\boldsymbol{\theta}_k) \prod_s q(\mathbf{z}_s), \quad (58)$$

where each factor takes its prior's exponential-family form: $q(\boldsymbol{\pi}_s) = \text{Dir}(\boldsymbol{\alpha}_s)$, $q(A_{s,k,\cdot}) = \text{Dir}(\boldsymbol{\alpha}_{s,k}^A)$, $q(\boldsymbol{\theta}_k) = \text{NIW}(\mu_k^q, \kappa_k^q, \Psi_k^q, \nu_k^q)$, and $q(\mathbf{z}_s)$ is computed by forward-backward.

Coordinate-ascent updates

State posterior $q(\mathbf{z}_s)$ via forward-backward. Given the current $q(\boldsymbol{\pi}_s)$, $q(A_s)$, $q(\boldsymbol{\theta})$, the optimal $q(\mathbf{z}_s)$ in the mean-field form is the standard HMM forward-backward marginals over states, with effective parameters that are the geometric expectations of the variational distributions:

$$\tilde{\pi}_{s,k} = \exp \mathbb{E}_q[\log \pi_{s,k}] = \exp(\psi(\alpha_{s,k}) - \psi(\sum_{k'} \alpha_{s,k'})), \quad (59)$$

$$\tilde{A}_{s,k,k'} = \exp \mathbb{E}_q[\log A_{s,k,k'}] = \exp(\psi(\alpha_{s,k,k'}^A) - \psi(\sum_j \alpha_{s,k,j}^A)), \quad (60)$$

$$\tilde{p}(y_{s,t} \mid z_{s,t} = k) = \exp \mathbb{E}_{q(\boldsymbol{\theta}_k)}[\log \mathcal{N}(y_{s,t}; \mu_k, \Sigma_k)], \quad (61)$$

where ψ is the digamma function. The forward-backward recursions then compute

$$\gamma_{s,t,k} := q(z_{s,t} = k \mid y_{s,1:T}), \quad (62)$$

$$\xi_{s,t,k,k'} := q(z_{s,t-1} = k, z_{s,t} = k' \mid y_{s,1:T}), \quad (63)$$

with $\gamma_{s,t,k}$ obtained from the standard $\alpha_t \beta_t / \sum_k \alpha_T(k)$ recursion using the geometric-mean parameters above.

Initial-state Dirichlet update.

$$\alpha_{s,k} = \alpha_{0,k} + \gamma_{s,1,k}. \quad (64)$$

Transition Dirichlet update.

$$\alpha_{s,k,k'}^A = \alpha_{A,k,k'} + \sum_{t=2}^T \xi_{s,t,k,k'}. \quad (65)$$

Emission NIW update. Define the expected sufficient statistics for state k , pooled across samples:

$$N_k = \sum_{s,t} \gamma_{s,t,k}, \quad (66)$$

$$\bar{y}_k = N_k^{-1} \sum_{s,t} \gamma_{s,t,k} y_{s,t}, \quad (67)$$

$$S_k = \sum_{s,t} \gamma_{s,t,k} (y_{s,t} - \bar{y}_k)(y_{s,t} - \bar{y}_k)^\top. \quad (68)$$

The NIW updates are then the standard conjugate form (Beal 2003, eqs. 3.65–3.68):

$$\kappa_k^q = \kappa_k^0 + N_k, \quad (69)$$

$$\mu_k^q = \frac{\kappa_k^0 \mu_k^0 + N_k \bar{y}_k}{\kappa_k^q}, \quad (70)$$

$$\nu_k^q = \nu_k^0 + N_k, \quad (71)$$

$$\Psi_k^q = \Psi_k^0 + S_k + \frac{\kappa_k^0 N_k}{\kappa_k^0 + N_k} (\bar{y}_k - \mu_k^0)(\bar{y}_k - \mu_k^0)^\top. \quad (72)$$

ELBO formula

The variational lower bound for the VB-HMM baseline is

$$\mathcal{L}_{\text{HMM}} = \mathbb{E}_q[\log p(\mathbf{y}, \mathbf{z}, \boldsymbol{\pi}, \mathbf{A}, \boldsymbol{\theta})] - \mathbb{E}_q[\log q(\mathbf{z}, \boldsymbol{\pi}, \mathbf{A}, \boldsymbol{\theta})], \quad (73)$$

which decomposes into the forward-backward log-marginal $\sum_s \log \sum_k \alpha_{s,T}(k)$ (using the geometric-mean parameters), minus the Dirichlet KLs on $\boldsymbol{\pi}_s$ and $A_{s,k,\cdot}$, minus the NIW KL on $\boldsymbol{\theta}_k$. Closed forms for each KL are standard.

Connection to Baum-Welch

Maximum-likelihood EM (Baum-Welch) is the degenerate case of this coordinate ascent where the priors are flat ($\alpha_0 = \alpha_A = 1$, NIW degenerating to a flat prior) and posteriors collapse onto delta functions at the maximum-likelihood parameters. FinaleMe uses Baum-Welch per sample; the VB-HMM baseline used here therefore strictly generalizes FinaleMe by retaining the priors and reporting variational posteriors rather than point estimates. This makes the comparison HAVI-Methyl vs. VB-HMM a like-for-like comparison of two principled variational methods, rather than a Bayesian-vs.-frequentist comparison that would conflate two different sources of advantage.

Limitations of the baseline

Even with the conjugate VB upgrade, the baseline retains FinaleMe’s structural limitations: binary state, no continuous $\beta \in [0, 1]$, per-sample fitting with no cross-sample pooling of transitions or emissions, and Markov-chain dependence on chromosomal neighbors that fails in CpG-poor regions. The HAVI-Methyl design replaces all four limitations directly, which is the substantive gap between the baseline and the proposed model.

C Reparameterization gradients

This appendix derives the reparameterization gradients used throughout HAVI-Methyl. We treat the Gaussian variational layers, the rational-quadratic spline of the neural-spline-flow encoder, the sticking-the-landing variant, and the doubly-reparameterized gradient (DReG) estimator for the IWAE bound.

Gaussian variational layers

For the population layer $q_{\lambda_\ell}(\mu_\ell^{\text{pop}}) = \mathcal{N}(m_\ell, v_\ell)$, the reparameterization is $\mu_\ell^{\text{pop}} = m_\ell + \sqrt{v_\ell} \xi$ with $\xi \sim \mathcal{N}(0, 1)$. The gradient of any expectation $g(\mu_\ell^{\text{pop}})$ with respect to the variational parameters is

$$\nabla_{m_\ell} \mathbb{E}[g] = \mathbb{E}[\nabla g], \quad \nabla_{v_\ell} \mathbb{E}[g] = \frac{1}{2\sqrt{v_\ell}} \mathbb{E}[\xi \nabla g], \quad (74)$$

both unbiased. The same form applies to the sample-shift layer with (m_s^δ, v_s^δ) .

Rational-quadratic spline (NSF) and its log-Jacobian

A neural-spline-flow block (Durkan et al., 2019) is a piecewise rational-quadratic transform $T : [-B, B] \rightarrow [-B, B]$ parameterized by K knots $\{(x_k, y_k)\}_{k=0}^K$ and $K-1$ interior derivatives $\{d_k\}_{k=1}^{K-1}$, with $d_0 = d_K = 1$ for tail behavior. Within bin k , with $\xi_k = (x - x_k)/(x_{k+1} - x_k)$ and $s_k = (y_{k+1} - y_k)/(x_{k+1} - x_k)$,

$$T(x) = y_k + \frac{(y_{k+1} - y_k)(s_k \xi_k^2 + d_k \xi_k (1 - \xi_k))}{s_k + (d_{k+1} + d_k - 2s_k) \xi_k (1 - \xi_k)}. \quad (75)$$

The log-Jacobian within bin k is

$$\log \frac{dT}{dx} = 2 \log s_k + \log(d_{k+1} \xi_k^2 + 2s_k \xi_k (1 - \xi_k) + d_k (1 - \xi_k)^2) - 2 \log(s_k + (d_{k+1} + d_k - 2s_k) \xi_k (1 - \xi_k)), \quad (76)$$

which is differentiable in (x_k, y_k, d_k) everywhere except at the bin boundaries (where the derivative is continuous by construction).

The full flow T_ϕ is a composition of K such blocks plus residual transformations. Its log-Jacobian is the sum over blocks of the per-block log-Jacobians. Both T_ϕ and $\log |\det \nabla T_\phi|$ are differentiable in ϕ and in the conditioning context c , so reparameterized gradients are computed by autodiff through the spline.

Sticking-the-landing

The reparameterization gradient of the negative ELBO with respect to encoder parameters ϕ contains a score-function term:

$$\nabla_\phi \text{KL}(q_\phi(\eta) \| p(\eta)) = \mathbb{E}_\epsilon [\nabla_\phi \log q_\phi(T_\phi(\epsilon)) + \nabla_\phi \log q_\phi(T_\phi(\epsilon)) \cdot (\log q_\phi(T_\phi(\epsilon)) - \log p(T_\phi(\epsilon)))]. \quad (77)$$

The first summand on the right has zero expectation (it is the score function), but nonzero variance. Sticking-the-landing (Roeder et al., 2017) replaces $\nabla_\phi \log q_\phi(T_\phi(\epsilon))$ by its evaluation at a stop-gradient copy of ϕ , which preserves the unbiased estimator while eliminating the variance contribution from this term. Empirically, the variance reduction is substantial when q_ϕ is close to $p(\eta | F)$ (i.e., late in training) and is the standard recommendation for reparameterized VAEs.

Doubly-reparameterized gradient (DReG) for IWAE

Rainforth et al. (2018) showed that the IWAE encoder gradient has signal-to-noise ratio that decreases as $K^{-1/2}$, where K is the number of importance samples, undermining the very benefit of using IWAE. Tucker et al. (2019) resolve this with the DReG estimator: instead of differentiating $\log \frac{1}{K} \sum_k w_k$ directly with respect to ϕ , they observe that

$$\nabla_\phi \log \left(\frac{1}{K} \sum_k w_k \right) = \sum_k \tilde{w}_k \nabla_\phi \log w_k, \quad \tilde{w}_k = \frac{w_k}{\sum_j w_j}, \quad (78)$$

and replace the inner $\nabla_\phi \log w_k$ by a doubly-reparameterized form with stop-gradient on the outer importance weights. This eliminates the score-function contribution at the per-sample level. Empirically DReG restores the SNR scaling and makes IWAE genuinely useful at $K \geq 8$.

Gradient-variance comparison

For small K (say $K \leq 4$), single-sample reparameterization with sticking-the-landing has the lowest variance and is the recommended setting during training. For larger K (during fine-tuning), DReG-IWAE has the lowest variance and tightens the bound. The HAVI-Methyl training recipe (Section 6) uses single-sample reparameterization at $K = 1$ during the main training run and DReG-IWAE at $K = 8$ during the fine-tuning pass.

D Architecture details

This appendix provides a detailed accounting of the HAVI-Methyl architecture, including parameter counts, dimensions, and the data flow from raw fragments to per-locus posterior. Table 6 enumerates every trainable component.

Table 6 HAVI-Methyl architecture. Parameter counts assume hidden dim $d_c = 128$, $L_e = 2$ ISAB layers, $K = 6$ NSF blocks with 8 spline bins, and $T = 64$ tissue components. Frozen modules (HyenaDNA, GoDMC anchors) are not counted.

Component	Specification	Output dim	Trainable params
<i>Fragment encoder</i>			
ISAB layer 1	4 heads, $m = 64$ inducing points, hidden 128	128	$\sim 100\text{K}$
ISAB layer 2	4 heads, $m = 64$ inducing points, hidden 128	128	$\sim 100\text{K}$
PMA pool ($k = 1$)	4 heads, hidden 128	128	$\sim 50\text{K}$
<i>Sequence encoder (one of)</i>			
Dilated CNN	6 layers, kernel 7, dilations $\{1, 2, 4, 8, 16, 32\}$	128	$\sim 200\text{K}$
HyenaDNA proj.	frozen 256-d $\rightarrow$ 128-d linear	128	$\sim 30\text{K}$
<i>Context fusion</i>			
Concat + MLP	$[c^{\text{frag}} \parallel c^{\text{seq}} \parallel m_\ell \parallel m_s^\delta] \rightarrow 128$	128	$\sim 70\text{K}$
<i>Normalizing flow posterior</i>			
NSF block $\times K = 6$	rational-quadratic, 8 bins, hidden 128	1 (logit- β)	$\sim 700\text{K}$
<i>Likelihood heads</i>			
Beta-Binomial sufficient-stat head	1 layer, dim 64	1 (n^m)	$\sim 10\text{K}$
End-motif categorical head	256-way softmax	256	$\sim 30\text{K}$
Coverage NB head	2-layer MLP, hidden 64	1 ($\log \mu$)	$\sim 10\text{K}$
<i>Tissue-of-origin head</i>			
Dirichlet ($T = 64$)	linear $\rightarrow T$, softplus	64	$\sim 10\text{K}$
<i>Variational parameters (non-amortized)</i>			
Population λ_ℓ	(m_ℓ, v_ℓ) , $\ell = 1, \dots, L \approx 30\text{M}$	$2L$	60M (data-scale)
Sample shift ν_s	(m_s^δ, v_s^δ) , $s = 1, \dots, S$	$2S$	$\sim S$
Total trainable encoder/heads			$\sim 1.0\text{M}$
Total variational parameters			$\sim 60\text{M}$

The encoder and heads are about 1 M trainable parameters, modest by deep-learning standards. The variational parameters dominate the parameter count, but they live on the wafer’s weight-streaming memory and are updated by closed-form natural-gradient updates rather than backprop through Adam, so the optimizer state is much smaller than for an equivalently-sized neural network.

Data flow

The end-to-end forward pass for one (s, ℓ) pair proceeds in stages. First, the per-fragment feature vectors $\{f_{s,\ell,j}\}_{j=1}^{n_{s,\ell}}$ are passed through two ISAB layers and pooled into a 128-dimensional $c_{s,\ell}^{\text{frag}}$. Second, the genome window ± 1 kb around locus ℓ is encoded into a 128-dimensional c_{ℓ}^{seq} by either the dilated CNN or the frozen HyenaDNA projection. Third, the current variational means m_{ℓ} (population) and m_s^{δ} (sample shift) are concatenated with the two encoder outputs and passed through a small MLP into a unified 128-dimensional context $c_{s,\ell}$. Fourth, $\epsilon \sim \mathcal{N}(0, 1)$ is reparameterized through the $K = 6$ NSF blocks conditioned on $c_{s,\ell}$ to produce $\eta_{s,\ell}$. Fifth, three likelihood heads are evaluated against the observed counts (Beta-Binomial), motifs (categorical), and coverage (NB), giving the per-locus reconstruction. Finally, the tissue-of-origin head consumes the per-sample posterior means across loci and produces a Dirichlet on tissue fractions.

Memory footprint

At $|B_s| = 8$ samples and $|B_{\ell}| = 4096$ loci per step, the dominant memory consumption is the ISAB intermediate activations: $|B_s| \cdot |B_{\ell}| \cdot \bar{n} \cdot d_c \cdot L_e \approx 8 \cdot 4096 \cdot 20 \cdot 128 \cdot 2 \approx 17$ MB per layer in float32, totaling about 35 MB across the encoder. The flow forward and Jacobian add another ~ 50 MB. Activation checkpointing on the encoder brings the peak below 100 MB per step, well within the 40 GB on-wafer SRAM of the wafer-scale accelerator.

E Synthetic simulator full specification

This appendix gives the full algorithmic specification of the chromatin-aware cfDNA simulator summarized in Section 10. The simulator is implemented in Python and released alongside the paper; the specification here is the formal description of its behavior, with each helper function defined explicitly together with its parameter table and the published source from which the parameters were drawn.

Inputs

The simulator takes four inputs: a reference genome assembly (hg38 in our setup); a chromatin-state annotation, taken from the 15-state Roadmap Epigenomics ChromHMM model and used to bias nucleosome positioning; a per-CpG ground-truth methylation track $\beta \in [0, 1]^L$ from a public reference atlas (we use Loyfer-2023 (Loyfer et al., 2023)); and a parameter dictionary controlling fragmentation, sequencing, and coverage.

Step 1: Nucleosome positioning

We sample nucleosome positions from a stationary one-dimensional point process with two ingredients. First, a renewal process with mean spacing 187 bp (the canonical inter-nucleosome distance in mammalian chromatin) and standard deviation 25 bp. Second, a Strauss process (Strauss, 1975) with interaction parameter $\gamma = 0.1$ that imposes nucleosome-nucleosome repulsion at distances below 147 bp (the nucleosome-DNA wrap length). The two combine to produce realistic positioning statistics that reproduce the canonical 200 bp peak in the empirical spacing distribution.

Step 2: Fragmentation cuts

Cut sites are sampled from a probability density biased toward linker DNA between nucleosomes:

$$p_{\text{cut}}(x) \propto \exp(-d_{\text{nuc}}(x)/\sigma_{\text{nuc}}) \cdot [1 + a \cos(2\pi d_{\text{nuc}}(x)/10.4)], \quad (79)$$

where $d_{\text{nuc}}(x)$ is the distance from x to the nearest nucleosome center, $\sigma_{\text{nuc}} = 20$ bp is the cut-site rolloff, $a = 0.3$ is the periodicity amplitude, and the 10.4 bp period reflects the rotational positioning of DNA around the nucleosome. The first term concentrates cut sites in linker DNA; the second imprints the canonical 10.4 bp periodicity observed in cfDNA fragment-length distributions.

Step 3: Fragment-length distribution

Fragment lengths L are sampled from a three-component Gaussian mixture corresponding to mono-, di-, and tri-nucleosomal fragments:

$$p(L) = 0.7\mathcal{N}(167, 15^2) + 0.2\mathcal{N}(332, 30^2) + 0.1\mathcal{N}(500, 80^2). \quad (80)$$

Parameters are drawn from [Snyder et al. \(2016\)](#), which characterizes plasma cfDNA fragment-length distributions across healthy individuals.

Step 4: End-motif sampling

The 4-mer end motif at each cut site is sampled from a categorical distribution over the 256 possible 4-mers, with logits conditioned on both the local genome sequence and the local methylation state:

$$m_{\text{cut}} \sim \text{Cat}(\text{softmax}(W_{\text{seq}} \text{onehot}(g_{\text{cut}}) + W_{\text{meth}} \beta_{\text{cut}})), \quad (81)$$

where g_{cut} is the four-base genome sequence at the cut site and β_{cut} is the local methylation. The motif logits are calibrated against the empirical 4-mer distribution from [Zhou et al. \(2022\)](#): the four most abundant motifs (CCCA, CCAG, CCTG, CCAA) account for approximately 20% of cuts in healthy plasma, and methylated cut sites show a $\sim 1.4\times$ enrichment for CG-containing motifs relative to unmethylated.

Step 5: Sequencing simulation

End-repair and adapter ligation are modeled by appending a fixed 5-bp adapter to each fragment end and applying a 1% per-base substitution error rate (matching the typical Illumina NovaSeq error profile). Paired-end 150 bp reads are emitted from each fragment, with read 1 aligned to the 5' end and read 2 to the 3' end. Fragments shorter than 150 bp emit overlapping read pairs.

Step 6: Coverage sampling

Per-locus fragment counts are sampled from a Negative-Binomial distribution conditioned on GC content and methylation:

$$n_{s,\ell} \sim \text{NegBin}(r = 2, p_{\ell}), \quad p_{\ell} = \text{logit}^{-1}(b_0 + b_{\text{GC}} \text{GC}_{\ell} + b_{\beta} \beta_{s,\ell}), \quad (82)$$

with coefficients $b_0, b_{\text{GC}}, b_{\beta}$ chosen to reproduce the empirical mean coverage and the GC-bias and methylation-bias regression coefficients reported in [Snyder et al. \(2016\)](#).

Table 7 Simulator parameter table with sources.

Parameter	Value	Source
Nucleosome spacing mean	187 bp	Empirical chromatin biology
Nucleosome spacing std	25 bp	Empirical chromatin biology
Strauss interaction γ	0.1	Strauss 1975
Cut-site rolloff σ_{nuc}	20 bp	Inferred from cfDNA
Periodicity amplitude a	0.3	Snyder et al. 2016
Periodicity period	10.4 bp	Helical pitch of B-DNA
Length mixture mode 1	$\mathcal{N}(167, 15^2)$	Snyder et al. 2016
Length mixture mode 2	$\mathcal{N}(332, 30^2)$	Snyder et al. 2016
Length mixture mode 3	$\mathcal{N}(500, 80^2)$	Snyder et al. 2016
Per-base error rate	1%	Illumina NovaSeq spec
Read length	150 bp paired-end	Illumina NovaSeq spec
NB dispersion r	2	Snyder et al. 2016

Parameter table

Computational cost

At $S = 80$ samples and $L = 10^5$ loci, simulation completes in approximately 30 minutes on a single CPU thread; full $L \approx 30\text{M}$ run is approximately 4 hours single-threaded but trivially parallelizes by chromosome. The released implementation includes a multiprocessing wrapper that scales linearly with thread count.

F Hyperparameter table

Symbol	Value	Description
<i>Model priors</i>		
μ_0, τ_0	0, 2.0	Population logit prior (broad)
σ_δ	0.5	Sample-shift std
σ_η	0.8	Per-(s, ℓ) latent std
κ	20	Beta-Binomial concentration
<i>Regularization weights</i>		
β_{VIB}	1.0	VIB weight
λ_{cf}	0.5	Counterfactual invariance weight
λ_{mQTL}	0.2	mQTL anchor weight
λ_{ToO}	1.0	Tissue-of-origin loss weight
<i>Architecture</i>		
d_c	128	Encoder hidden dim
L_e	2	ISAB layers
m	64	ISAB inducing points
K flow blocks	6	NSF stack depth
NSF bins	8	Spline knots per block
T_{max}	64	HDP truncation level
<i>Optimization</i>		
LR encoder	10^{-3}	AdamW
Weight decay	10^{-4}	AdamW
ρ_t	$(t+1)^{-0.6}$	Robbins-Monro for natural gradients
$ B_s $	8	Sample mini-batch size
$ B_\ell $	4096	Locus mini-batch size
g_{clip}	5.0	Gradient clip global norm
T_{anneal}	1 epoch	KL annealing schedule
LR warmup	5% of steps	Linear warmup
LR decay	cosine	Post-warmup
<i>IWAE / fine-tuning</i>		
$K_{\text{IWAE}}^{\text{train}}$	1	Single-sample reparam during main training
$K_{\text{IWAE}}^{\text{finetune}}$	8	DReG-IWAE during fine-tuning
<i>Data</i>		
Train/val/test split	70/15/15	By individual
Held-out chromosomes	{1, 22}	For chromosome-stratified test

G Notation glossary

Symbol	Type	Meaning	Introduced
<i>Indices</i>			
s	$\in \{1, \dots, S\}$	Sample index	§3
ℓ	$\in \{1, \dots, L\}$	Locus index	§3
j	$\in \{1, \dots, n_{s,\ell}\}$	Fragment index within (s, ℓ)	§3
k	$\in \{m, u\}$ or $\{1, \dots, T\}$	State / tissue index	§B, §9
<i>Latents</i>			
$\beta_{s,\ell}$	$\in (0, 1)$	Methylation β -value	§3
$\eta_{s,\ell}$	$\in \mathbb{R}$	Logit-methylation, $\text{logit}\beta_{s,\ell}$	§3
μ_ℓ^{pop}	$\in \mathbb{R}$	Population logit at locus ℓ	§3
δ_s	$\in \mathbb{R}$	Per-sample logit shift	§3
π_s	$\in \Delta^{T-1}$	Tissue fractions for sample s	§9
ζ	$\in \mathbb{R}^{d_c}$	Encoder representation	§7
U	latent	Unmeasured confounders	§7
Z	multi-component	All latents (App. A only)	§A
<i>Observations</i>			
$F_{s,\ell}$	bag of f_j	Fragment-feature bag at locus ℓ	§3
$f_{s,\ell,j}$	$\in \mathbb{R}^d$	Single fragment feature vector	§3
$n_{s,\ell}$	$\in \mathbb{Z}_{\geq 0}$	Fragment count at locus	§3
$n_{s,\ell}^{\text{cpg}}$	$\in \mathbb{Z}_{\geq 0}$	CpG-trial count at locus	§3
$n_{s,\ell}^m$	$\in \mathbb{Z}_{\geq 0}$	Methylated CpG-call count	§3
$m_{s,\ell,j}$	$\in \{1, \dots, 256\}$	4-mer end-motif of fragment j	§3
$X_{\text{prior},s}$	buffy-coat	Confound input for sample s	§7
g_ℓ	$\in \{0, 1, 2\}$	mQTL genotype at locus ℓ	§7
$y_{s,t}$	$\in \mathbb{R}^d$	Per-locus features for HMM baseline	§B
<i>Variational parameters</i>			
$\lambda_\ell = (m_\ell, v_\ell)$	$\mathbb{R}^2$	Population variational params	§4
$\nu_s = (m_s^\delta, v_s^\delta)$	$\mathbb{R}^2$	Sample-shift variational params	§4
ϕ	$\mathbb{R}^P$	Encoder + flow parameters	§4
T_ϕ	flow	Normalizing-flow transform	§4
$c_{s,\ell}$	$\mathbb{R}^{d_c}$	Encoder context vector	§4
<i>Model hyperparameters</i>			
μ_0, τ_0	$\mathbb{R}, \mathbb{R}_+$	Population prior mean and std	§3
$\sigma_\delta, \sigma_\eta$	$\mathbb{R}_+$	Per-sample, per-cell stds	§3
κ	$\mathbb{R}_+$	Beta-Binomial concentration	§3
β_{VIB}	$\mathbb{R}_+$	VIB Lagrangian weight	§7
$\lambda_{\text{cf}}, \lambda_{\text{mQTL}}, \lambda_{\text{ToO}}$	$\mathbb{R}_+$	Loss weights	§9
<i>Mathematical operators</i>			
$\sigma(\cdot)$	$\mathbb{R} \rightarrow (0, 1)$	Logistic sigmoid	—
$\text{logit}(\cdot)$	$(0, 1) \rightarrow \mathbb{R}$	Logit, $\log \frac{x}{1-x}$	—
$\mathcal{L}$	$\mathbb{R}$	Evidence lower bound	§5
$\text{KL}(p q)$	$\mathbb{R}_{\geq 0}$	Kullback-Leibler divergence	—
$\mathbb{E}_q[\cdot]$	operator	Expectation under q	—

H Simulator validation

This appendix reports validation of the cfDNA simulator described in Section 10 and Appendix E against published cfDNA distributions on three axes.

Fragment-length distribution

The simulated fragment-length distribution should reproduce the canonical mononucleosomal mode at 167 bp and secondary modes at 332 bp and ~ 500 bp. We verified this by sampling 10^5 fragments and comparing the empirical length histogram to the published distribution from Snyder et al. (2016), Figure 1. The simulated distribution exhibits the dominant peak at 167 bp (height 0.018 per bp), the secondary peak at 332 bp (height 0.0050), the tertiary peak near 500 bp (height 0.0009), and Kolmogorov-Smirnov distance of 0.04 to the published reference.

End-motif frequencies

We verified that the simulated end-motif frequencies at the 5' end match published plasma WGS distributions. The four most abundant simulated 4-mers (CCCA, CCAG, CCTG, CCAA) account for 19.2% of cuts, in good agreement with the $\sim 20\%$ figure reported in healthy plasma in Zhou et al. (2022), Table 1. The Kullback-Leibler divergence between the simulated 4-mer distribution and the empirical distribution is 0.08 nats.

Periodicity spectrum

The autocorrelation of cut sites should exhibit the canonical 10.4 bp periodicity peak. We verified this by computing the autocorrelation of cut indicators across 10 kb windows and finding the dominant peak at 10.42 bp, ± 0.04 bp, with secondary peaks at integer multiples consistent with the rotational positioning model.

Methylation-conditioned cut bias

A characteristic feature of cfDNA fragmentation is that methylated regions show different cut-site preferences from unmethylated regions, producing the methylation-fragmentomics correlation that FinaleMe and HAVI-Methyl exploit. We verified that simulated methylated regions show a $1.4\times$ enrichment of CG-containing motifs at cut sites relative to unmethylated regions, consistent with the published methylation-fragmentomics literature.

I Extended results

This appendix collects extended synthetic-experiment numbers, sensitivity analysis, and ablation tables that complement the main text. All numbers are produced from the released code at random seed 20260429 unless otherwise stated.

Per-coverage detail

Identifiability ablation

The two-decade reduction across the three regimes is consistent with Proposition 13.2 (positive identification under IV anchors) and Proposition 13.3 with Corollary 13.4 (asymptotic VIB leakage bound with finite- β_{VIB} refinement).

Tissue-fraction recovery

The 17-fold improvement reflects (i) the cost of binarization in the baseline pipeline (continuous- β inputs preserve graded methylation patterns), (ii) end-to-end joint training of the deconvolution

Table 8 Extended synthetic recovery results. Pearson r , RMSE, MAE, and 90%-credible-interval coverage and width across coverage levels.

Coverage	Method	Pearson r	RMSE	MAE	90% cov	90% width	CpG-poor r
0.1×	FinaleMe-style	0.182	0.309	0.270	0.873	0.865	0.224
0.1×	HAVI-Methyl	0.382	0.296	0.263	0.366	0.422	0.450
1×	FinaleMe-style	0.528	0.309	0.248	0.487	0.431	0.588
1×	HAVI-Methyl	0.829	0.220	0.194	0.449	0.371	0.870
5×	FinaleMe-style	0.779	0.184	0.141	0.658	0.362	0.833
5×	HAVI-Methyl	0.895	0.132	0.107	0.646	0.259	0.927
30×	FinaleMe-style	0.961	0.097	0.081	0.554	0.183	0.978
30×	HAVI-Methyl	0.966	0.096	0.081	0.403	0.126	0.981

Table 9 Prior-leakage attribution under three regularization regimes. Lower is better.

Configuration	Partial R^2 of prior input
No regularization (VIB off, mQTL off)	3.32%
VIB only ($\beta_{\text{VIB}} = 0.3$)	0.36%
VIB + mQTL anchors	0.022%

head against the methylation likelihood, and (iii) propagation of per-CpG posterior uncertainty into the tissue-fraction posterior via the integrated likelihood form of Section 9.

Sensitivity analysis

We swept the three model hyperparameters $(\kappa, \sigma_\eta, \sigma_\delta)$ at 5× coverage to characterize sensitivity. Pearson r varies by less than ± 0.02 across the range $\kappa \in [10, 40]$, $\sigma_\eta \in [0.5, 1.2]$, $\sigma_\delta \in [0.2, 1.0]$, indicating the model is not finely tuned to the default settings. The largest sensitivity is to κ (Beta-Binomial concentration) at very low coverage (0.1×), where small κ produces overdispersed predictions that worsen RMSE; the default $\kappa = 20$ is comfortably in the well-performing region.

Table 10 Tissue-fraction RMSE on synthetic three-tissue mixtures. Lower is better.

Method	Tissue-fraction RMSE
FinaleMe-binarized + QP deconvolution	0.153
HAVI-Methyl joint Dirichlet head	0.009